
Chapter 10

Stability and Frequency Compensation

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Outline

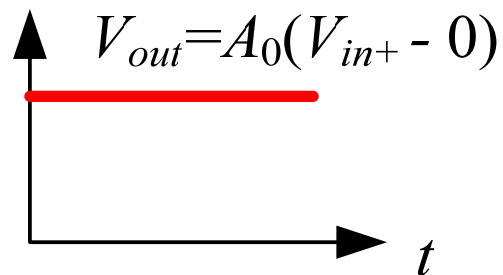
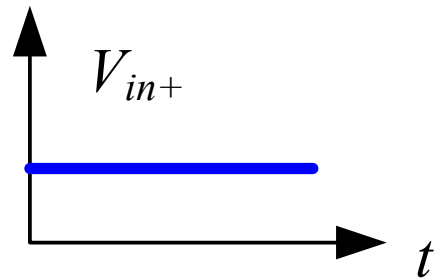
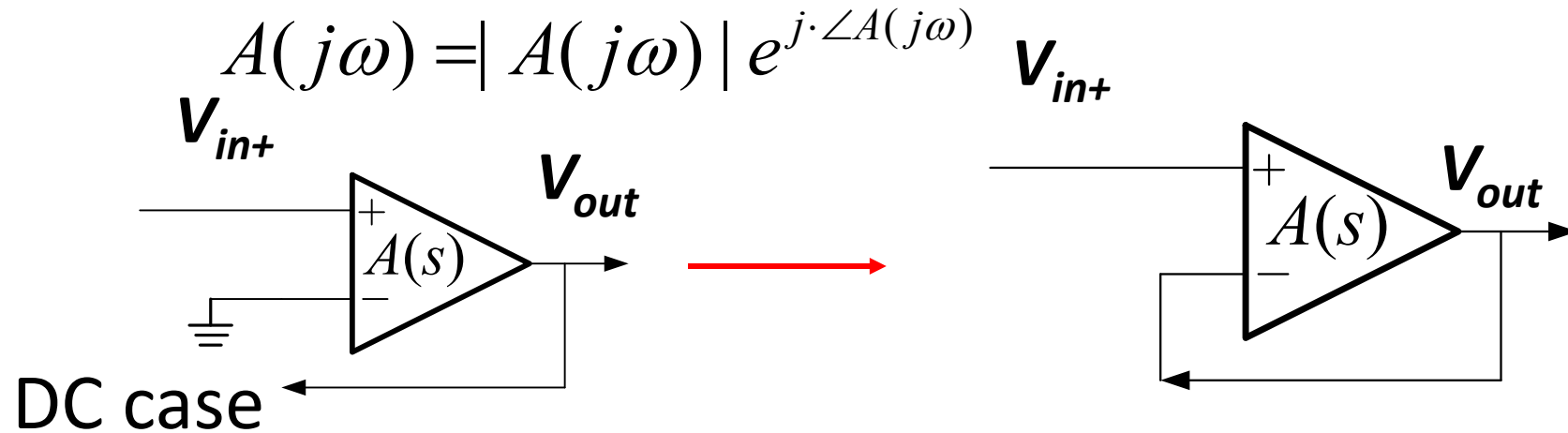
- **General consideration**
- Multipole systems
- Phase Margin
- Frequency compensation
- Compensation of two-stage OTAs
- Other compensation techniques



Stability issues in negative feedback systems

- **Advantages of negative feedback**
 - Gain desensitization
 - Imp. modification (by a factor of $(1+\text{loop gain})$)
 - Bandwidth extension
 - Nonlinearity reduction
- **Issues of negative feedback**
 - Instability
 - Tends to oscillate

Understanding of instability in time domain



$$V_{out} = A_0 (V_{in+} - V_{out})$$

$$V_{out} = \frac{A_0}{1 + A_0} V_{in+}$$

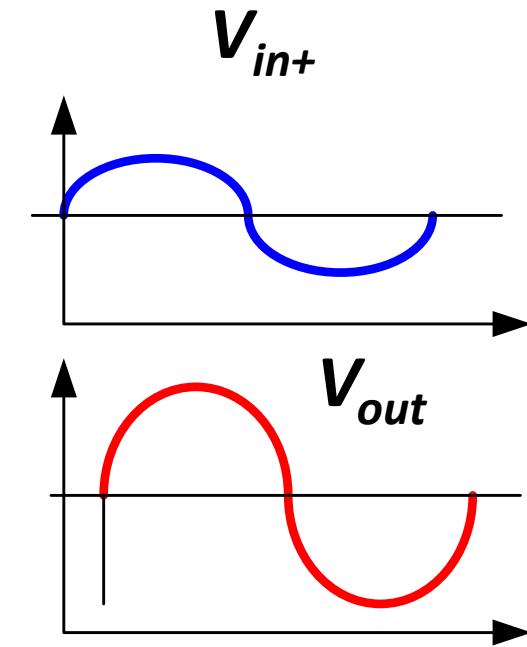
$$\rightarrow V_{in+} - V_{in-} = \frac{1}{1 + A_0} V_{in+}$$



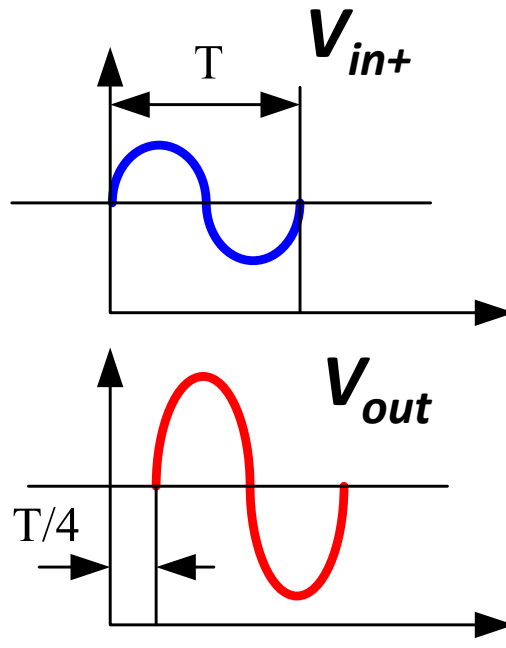
Sinusoid Case

$$A(j\omega) = |A(j\omega)| e^{j \cdot \angle A(j\omega)}$$

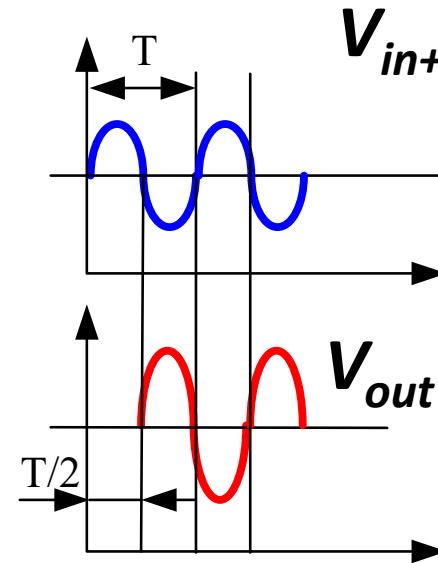
Frequency increases



$$\angle A(j\omega) \approx 0$$



$$\angle A(j\omega) \approx \frac{\pi}{2}$$

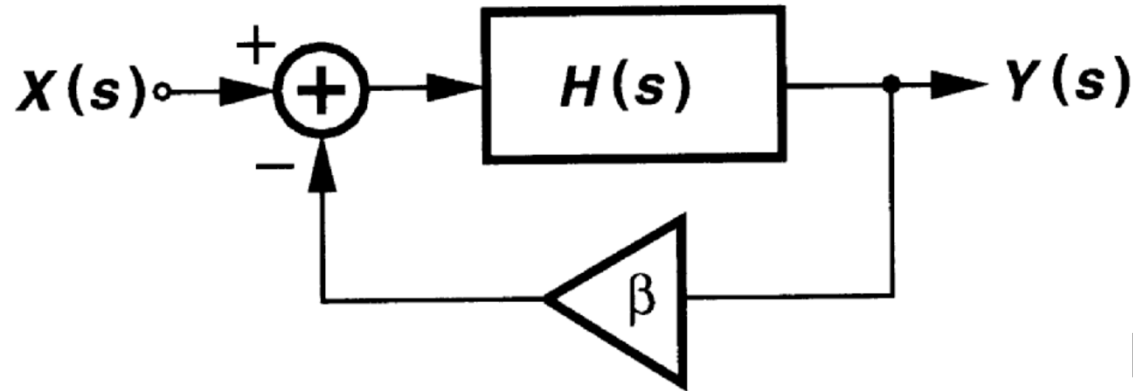


$$\angle A(j\omega) \approx \pi$$

Positive feedback!!



Barkhausen criteria



Loop gain:

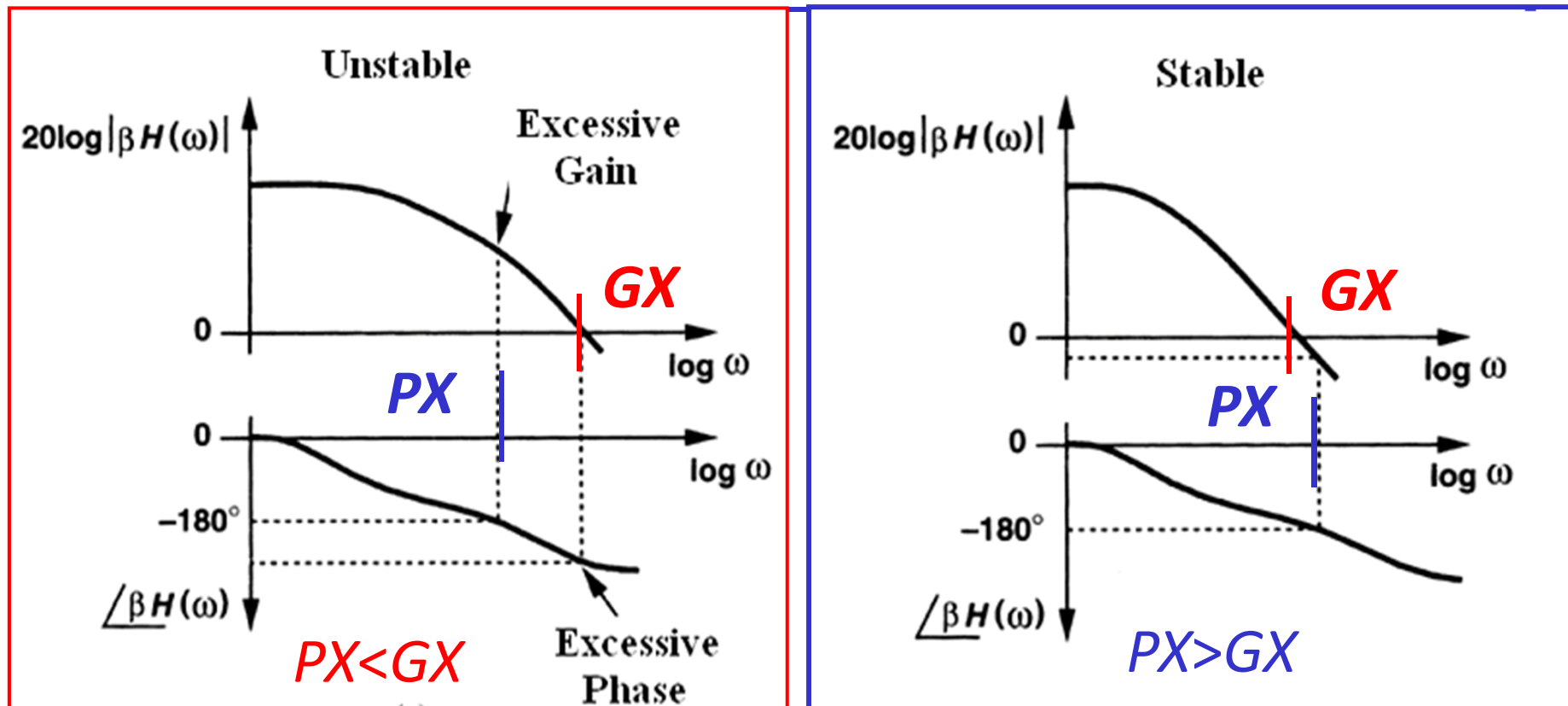
$$\frac{Y}{X}(s) = \frac{H(s)}{1 + \beta H(s)}$$

$$\begin{aligned} \beta H(j\omega) \\ = re^{j\varphi} \end{aligned}$$

May oscillate at ω_1 , if $|\beta H(j\omega_1)| = 1$ and $\angle \beta H(j\omega_1) = -180^\circ$
(*Barkhausen criteria*)



Stable and Unstable Systems



Stable system:

When gain is 0dB, phase is less than 180°

When phase is 180° , gain is less than 0dB

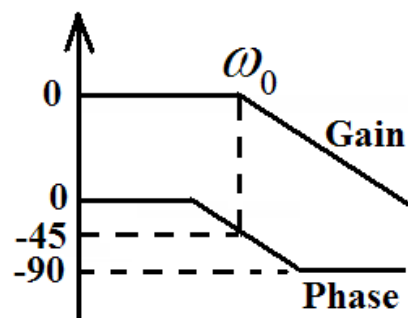


Review: Rules of Bode Plot

LHP: $-\omega_0$

$$\frac{1}{1 + s / \omega_0}$$

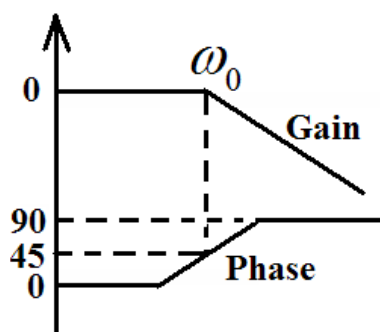
$$\frac{1}{1 + j(\omega / \omega_0)}$$



RHP: ω_0

$$\frac{1}{1 - s / \omega_0}$$

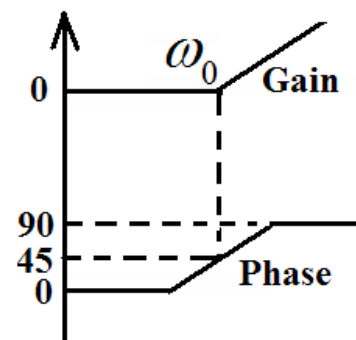
$$\frac{1}{1 - j(\omega / \omega_0)}$$



LHZ: $-\omega_0$

$$1 + s / \omega_0$$

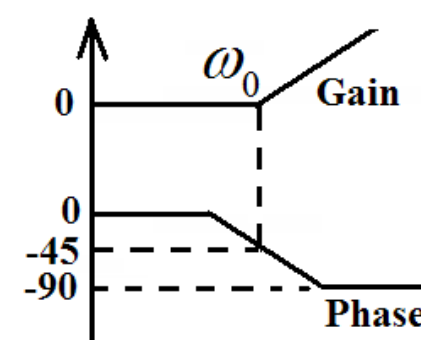
$$1 + j(\omega / \omega_0)$$



RHZ: ω_0

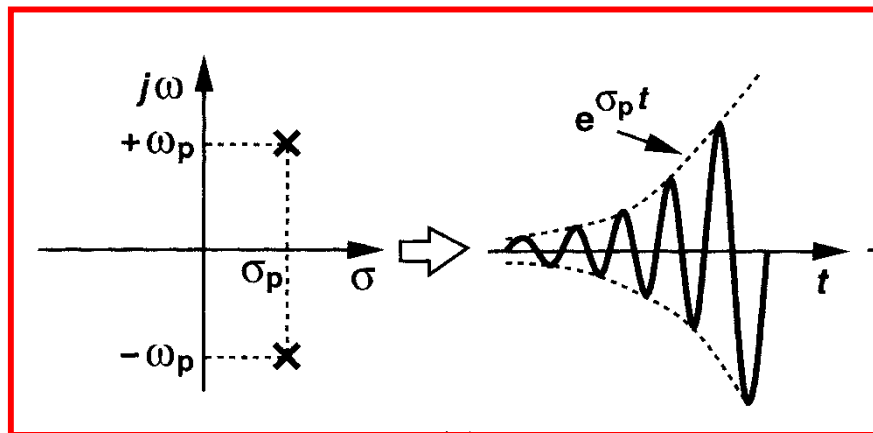
$$1 - s / \omega_0$$

$$1 - j(\omega / \omega_0)$$

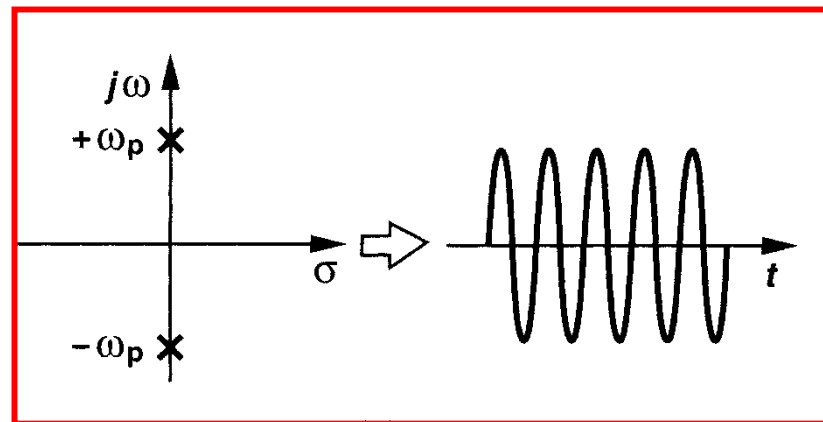


Stability and Complex Poles of Closed-Loop

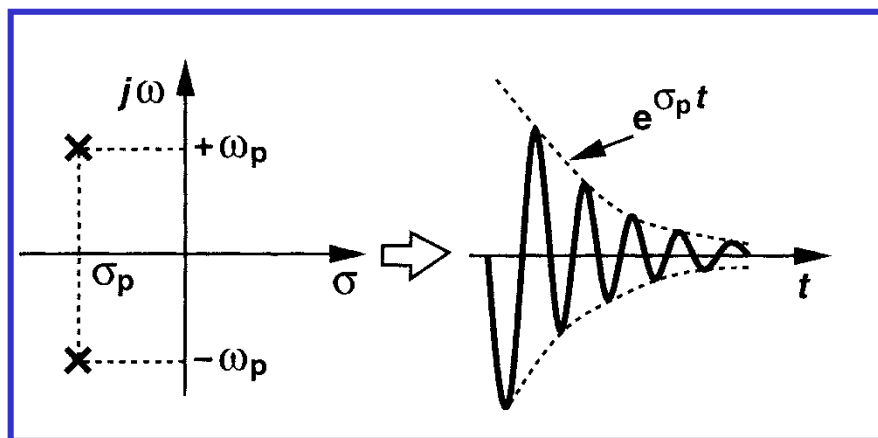
Impulse response



RHP poles



Pure imaginary poles

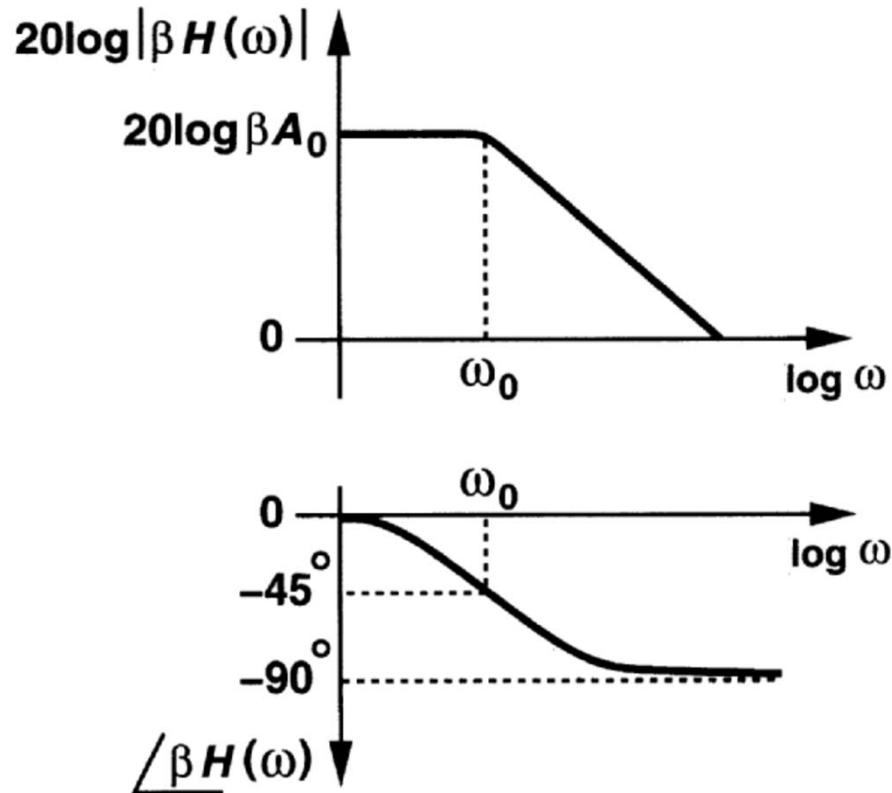


LHP poles

Stable

Basic Single-Pole System: Bode Plot

$$H(s) = A_0 \frac{1}{1 + \frac{s}{\omega_0}}$$



The Largest phase shift is 90°

Unconditional stable

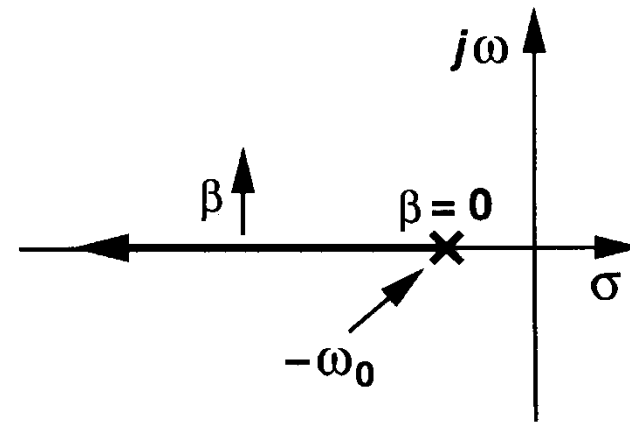


EX10. 1

- Root locus of one-pole system

$$H(s) = A_0 \frac{1}{1 + \frac{s}{\omega_0}}$$

$$\frac{Y(s)}{X(s)} = \frac{H(s)}{1 + \beta H(s)} = \frac{\frac{A_0}{1 + \beta A_0}}{1 + \frac{s}{(1 + \beta A_0)\omega_0}}$$



- The closed-loop pole is always on LHP
 - Unconditional stable

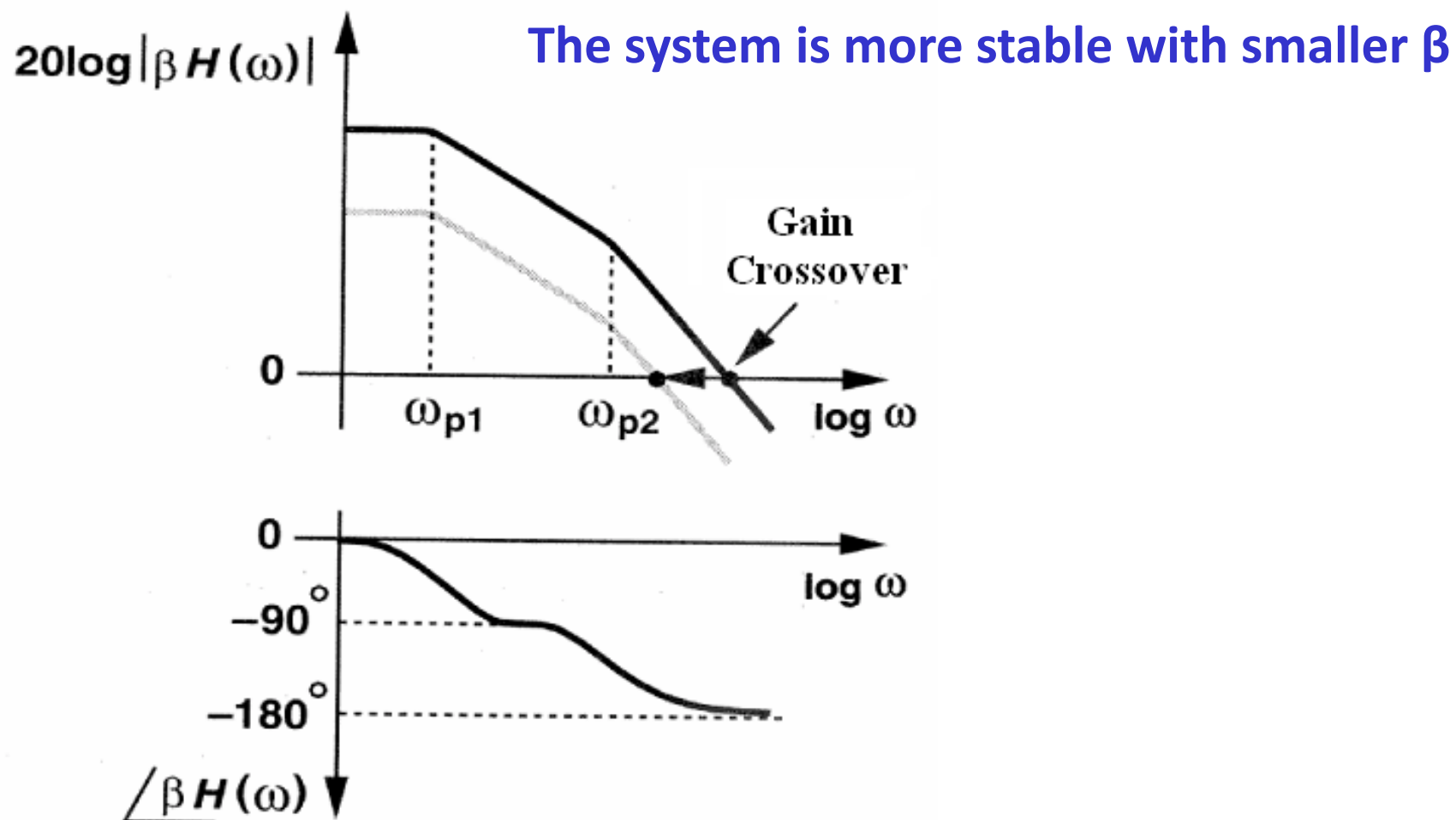


Outline

- General consideration
- **Multipole systems**
- Phase margin
- Frequency compensation
- Compensation of two-stage OTAs
- Other compensation techniques



Two-pole Systems

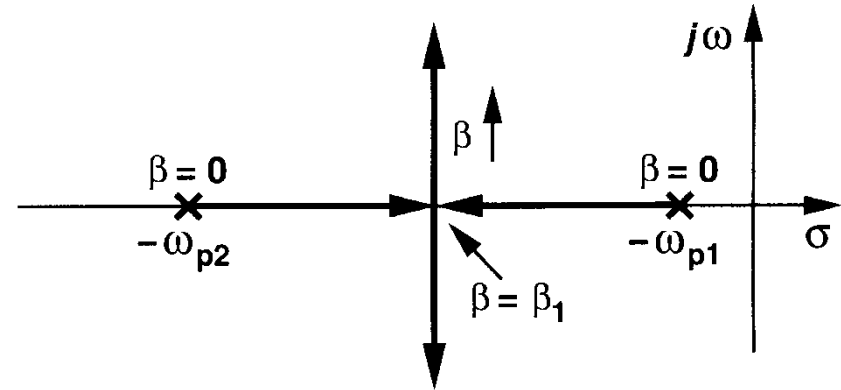


The worst situation at $\beta=1$

Ex. 10.3: Root-locus of two-pole system

- Open-loop transfer function

$$H(s) = \frac{A_0}{(1 + s / \omega_{p1})(1 + s / \omega_{p2})}$$



- Closed-loop transfer function

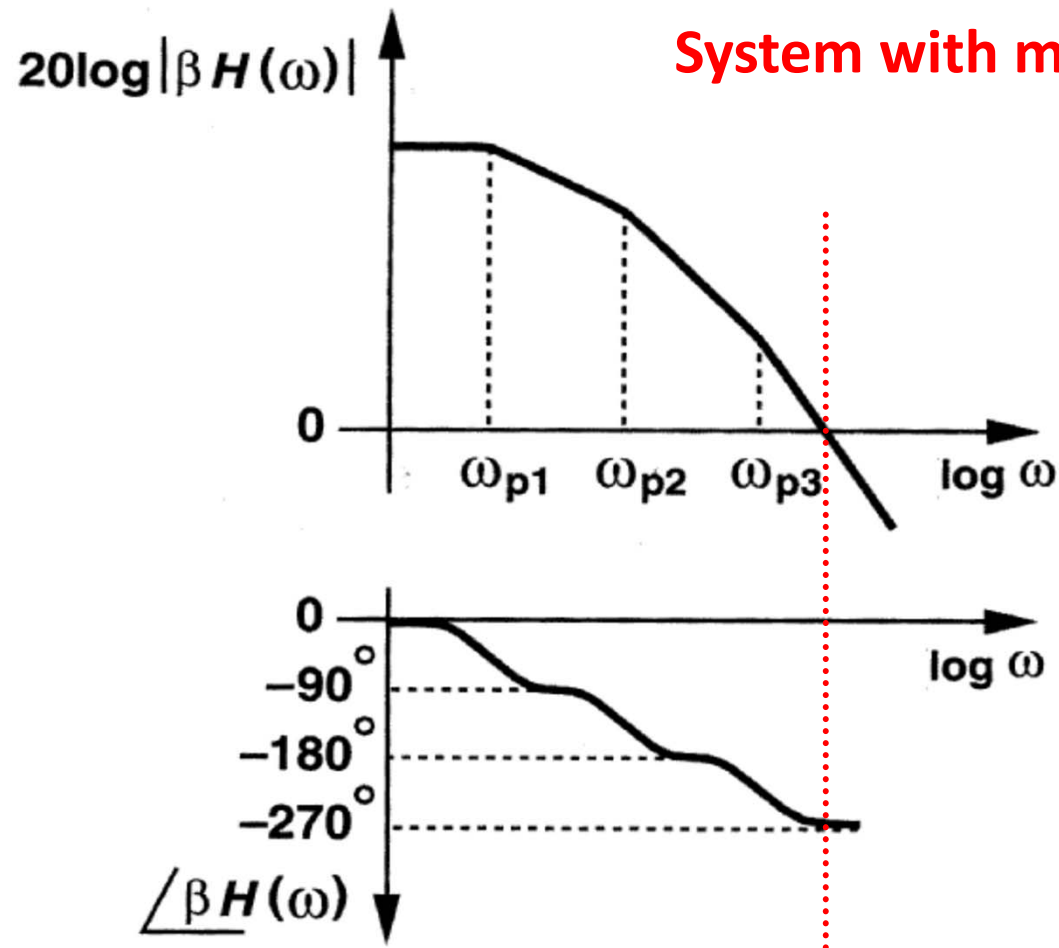
$$\frac{Y(s)}{X(s)} = \frac{A_0}{(1 + s / \omega_{p1})(1 + s / \omega_{p2}) + \beta A_0}$$

$$s_{1,2} = \frac{-(\omega_{p1} + \omega_{p2}) \pm \sqrt{(\omega_{p1} + \omega_{p2})^2 - 4(1 + \beta A_0)\omega_{p1}\omega_{p2}}}{2}$$

- Two-pole system is always stable



Three-Pole Systems



System with more than 3 poles tends to be unstable !!



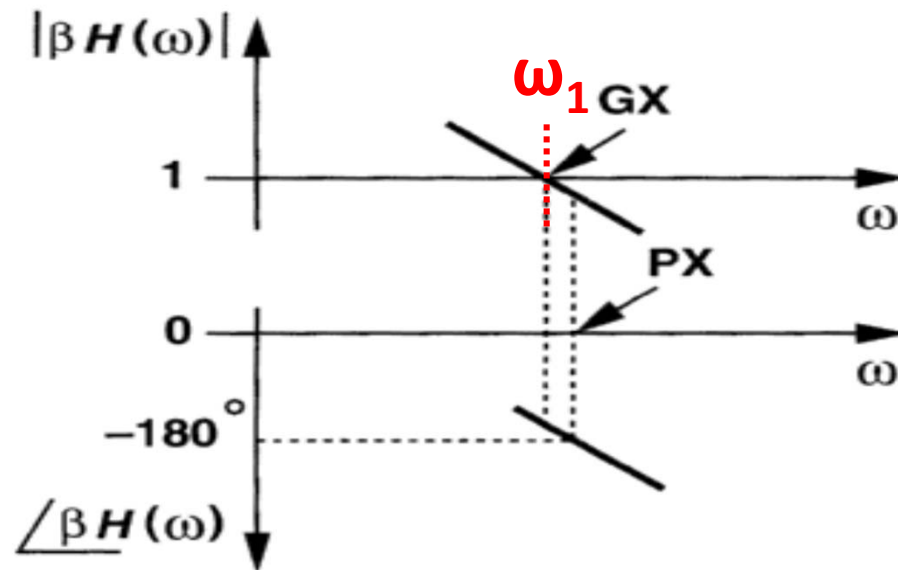
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Phase Margin

- For stable systems:
 - $|\beta H|$ must drop to 1 before phase cross -180°
 - How far should PX be from the GX?



Phase margin is defined as

$$PM = 180^\circ + \angle \beta H \Big|_{\omega=\omega_1}$$



Example (1): PM = 5°

$$GX=\omega_1, |\beta H(j\omega_1)|=1, \angle \beta H(j\omega_1) = -175^\circ.$$

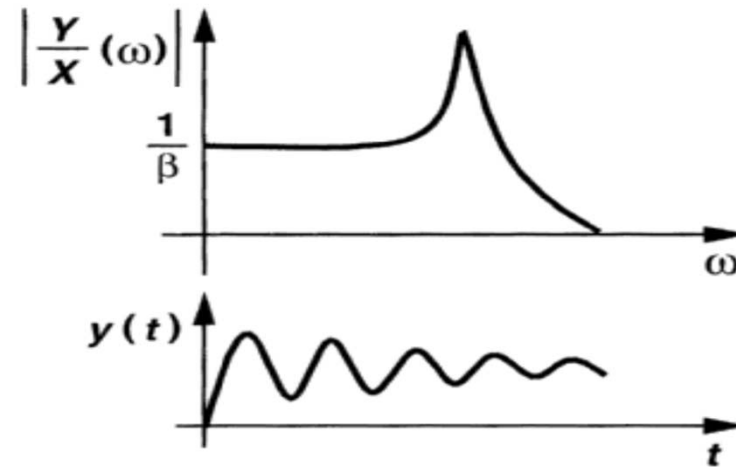
Here PM is 5°

$$\left| \frac{Y}{X}(j\omega_1) \right| = \frac{1}{\beta} \frac{1}{0.0872} \approx \frac{11.5}{\beta}$$

$$\frac{Y}{X}(j\omega_1) = \frac{H(j\omega_1)}{1 + \beta H(j\omega_1)}$$

$$= \frac{\frac{1}{\beta} \exp(-j175^\circ)}{1 + \exp(-j175^\circ)}$$

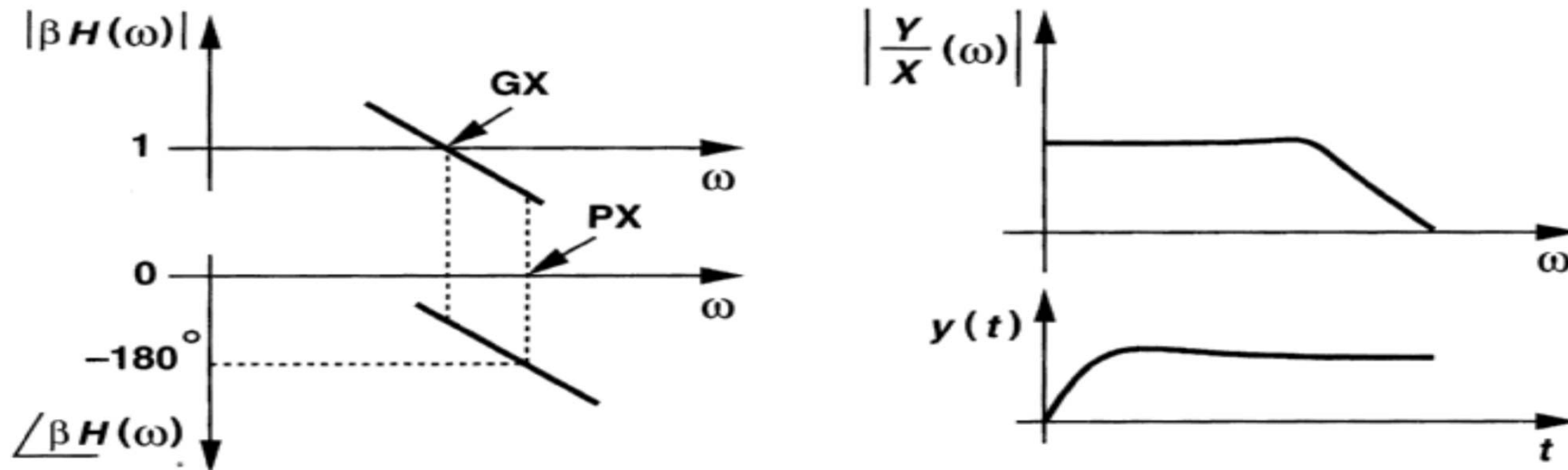
$$= \frac{1}{\beta} \cdot \frac{-0.9962 - j0.0872}{0.0038 - j0.0872}$$



Output suffers from ringing
Settling is slow down



Example (2): Large enough PM

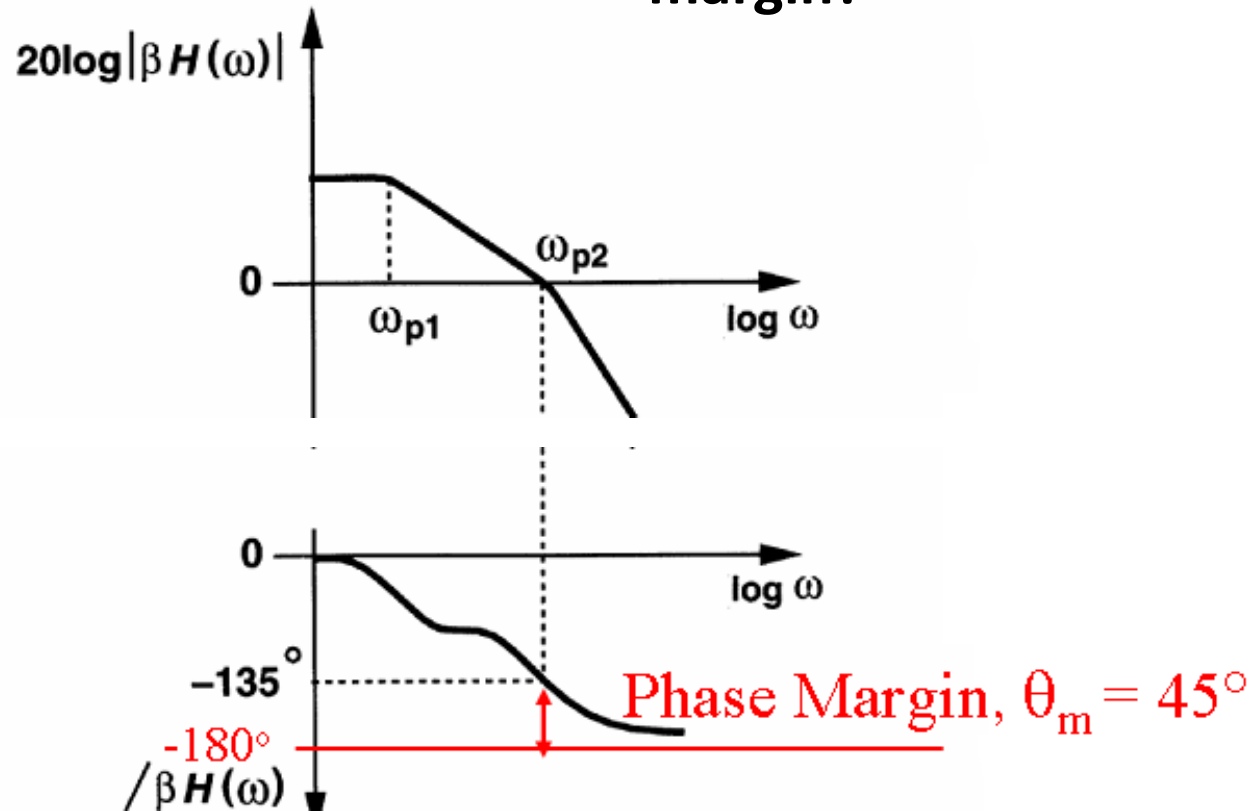


When the PM is large enough:

The overshoot in frequency response and ringing in the step response disappears

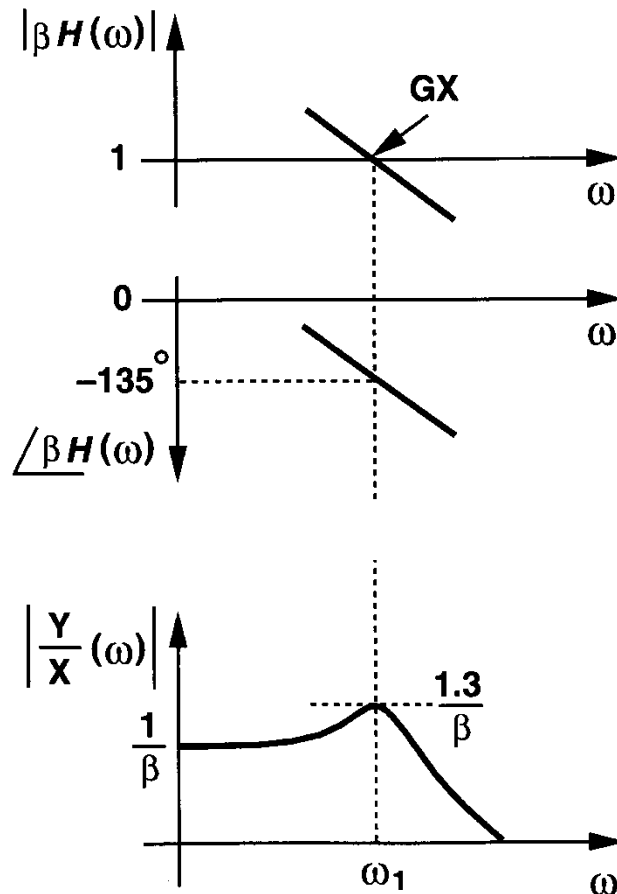
Ex. 10.4: Estimate the PM

$|\beta H(\omega_{p2})|=1$ and $|\omega_{p1}| \ll |\omega_{p2}|$, how much is the Phase margin?



How much PM is adequate?

- For $PM=45^\circ$



$$\frac{Y}{X}(j\omega_1) = \frac{H(j\omega_1)}{1 + 1 \times \exp(-j135^\circ)}$$

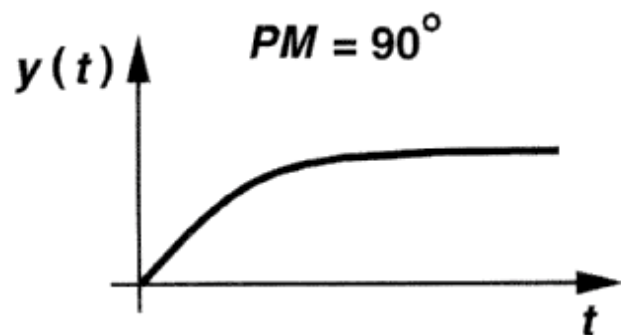
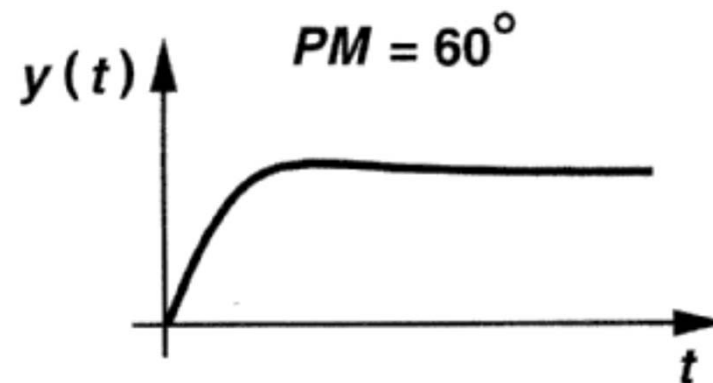
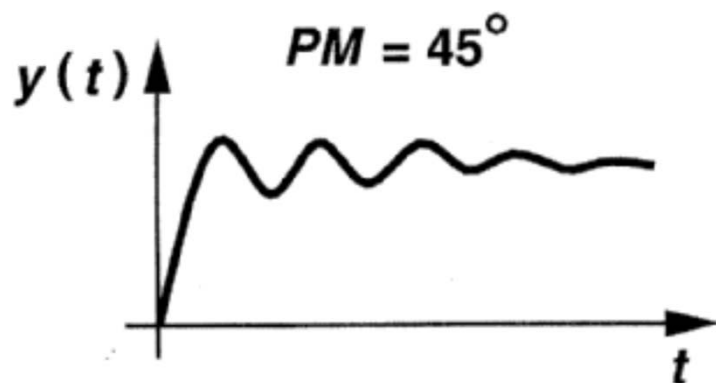
$$\frac{Y}{X}(j\omega_1) = \frac{H(j\omega_1)}{0.29 - 0.71j}$$

$$\left| \frac{Y}{X} \right|_{j\omega_1} = \frac{1}{\beta} \cdot \frac{1}{|0.29 - 0.71j|} \approx \frac{1.3}{\beta}$$

**The Peaking is not acceptable
for most applications**



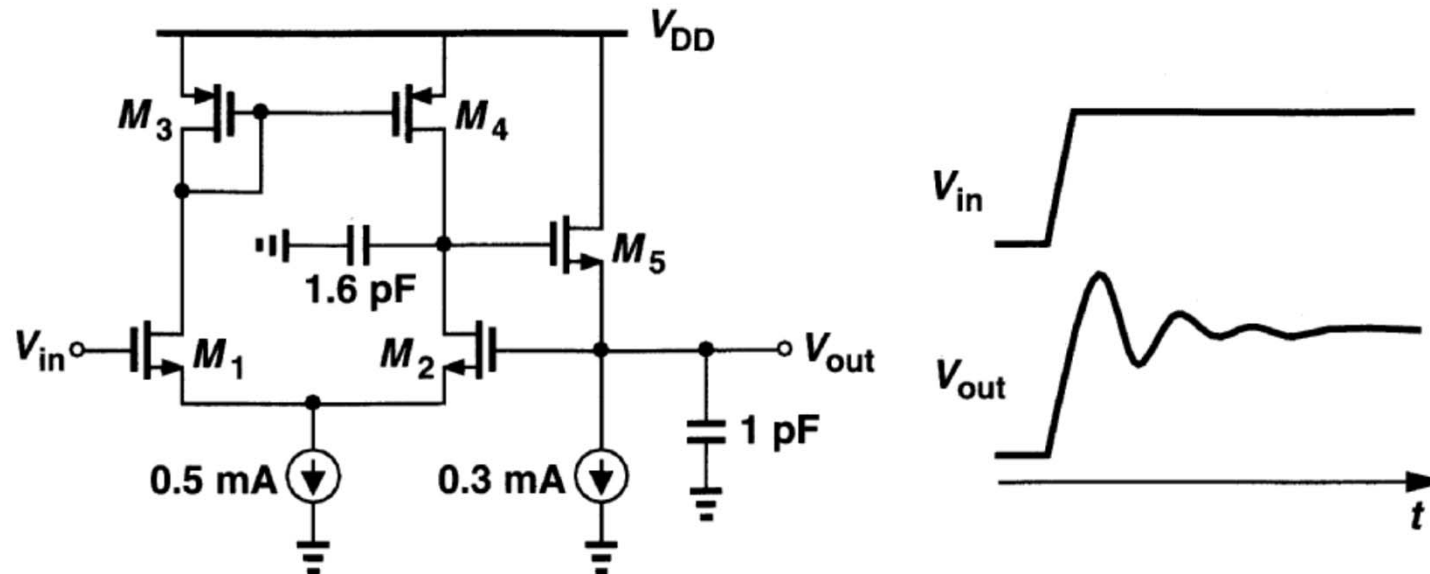
Time response of different PM



Best PM is between 60° and 70°

Small Signal Analysis Limitations

PM=65°



Small signal analysis yields 65° of θ_m , but **large signal** transient response is different. Large signal simulation includes effects not seen in small signal analysis.



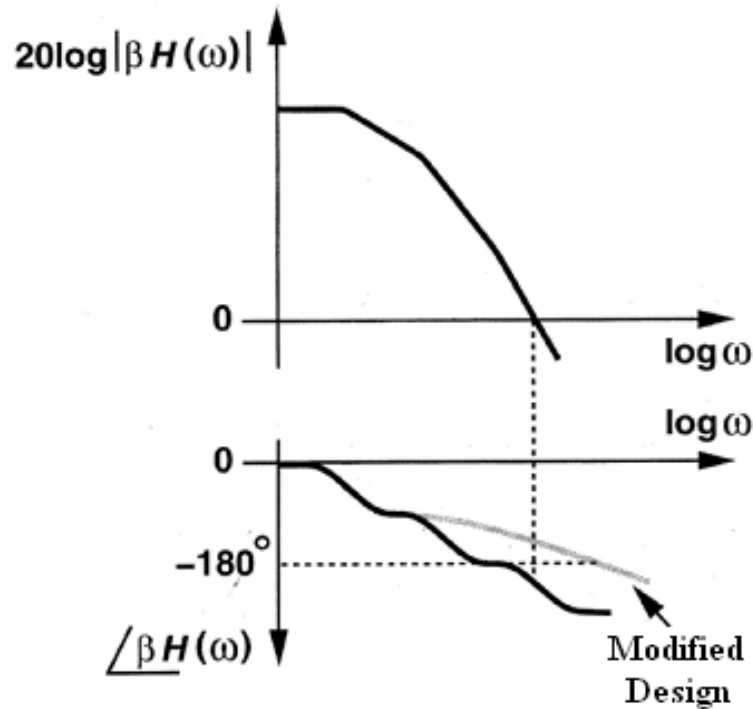
Outline

- General consideration
- Multipole systems
- Phase Margin
- **Frequency compensation**
 - **Single-ended telescopic OTA**
 - Fully differential telescopic OTA
- Compensation of two-stage OTAs
- Other compensation techniques



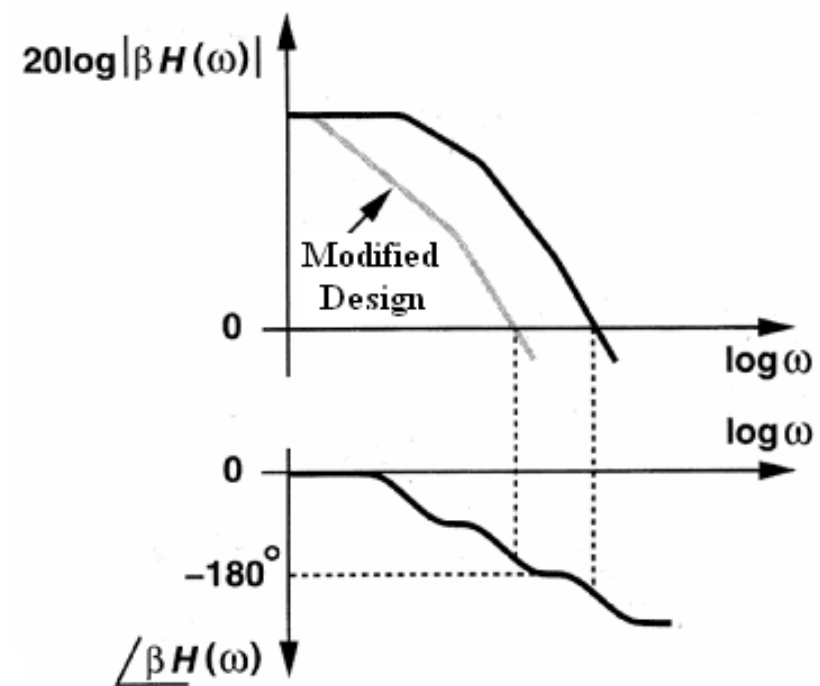
Methods of frequency compensation

Moving PX out



Minimize number of poles

Pushing GX in

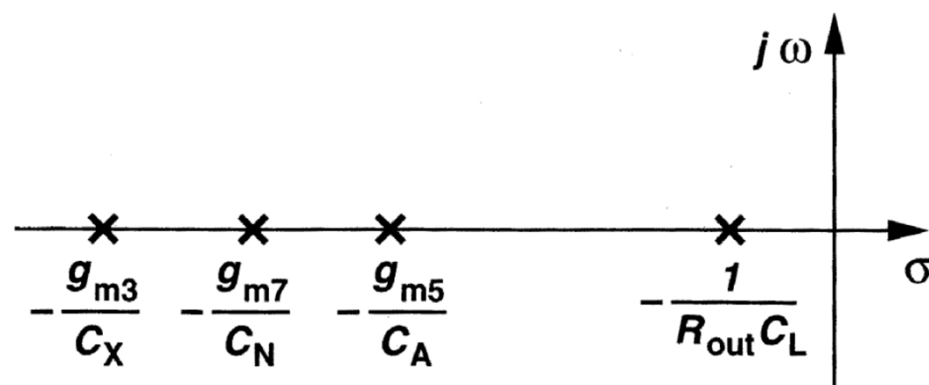
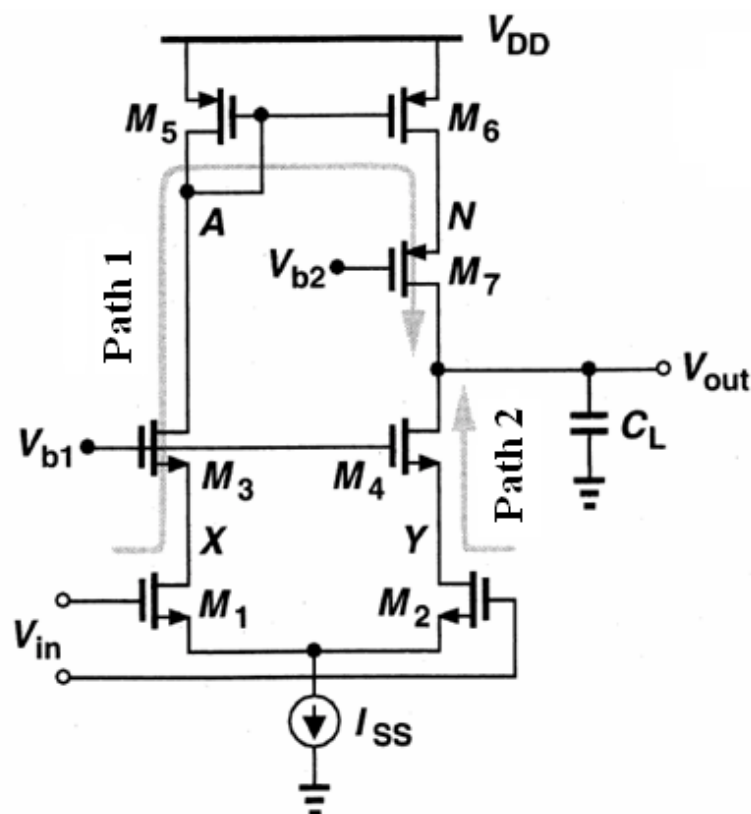


Move the dominant pole in



Single Ended Single Stage Amp

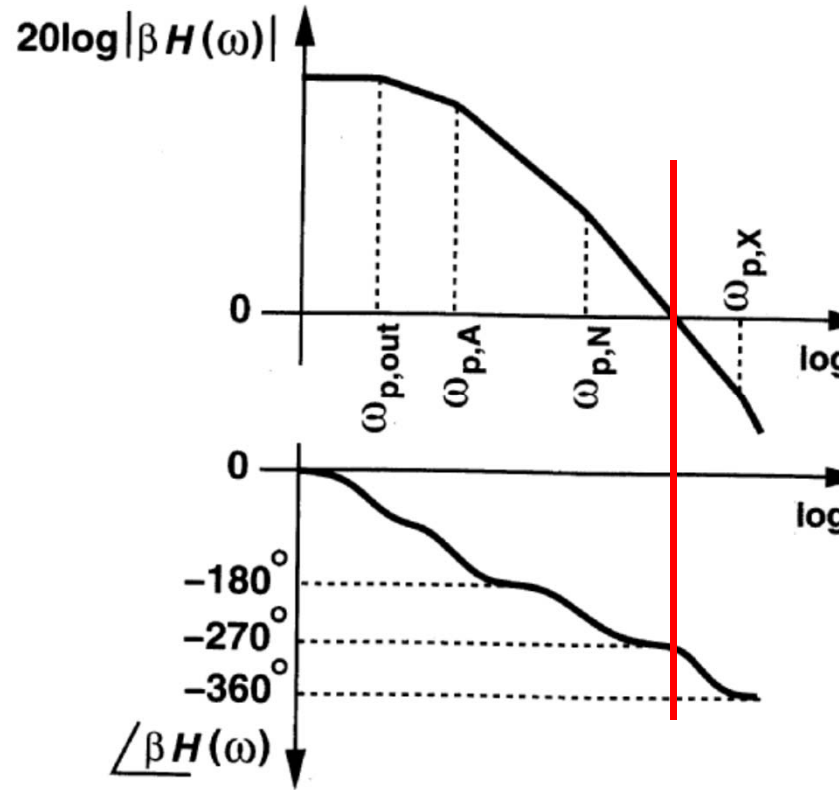
Dominant pole is at the output



Pole locations

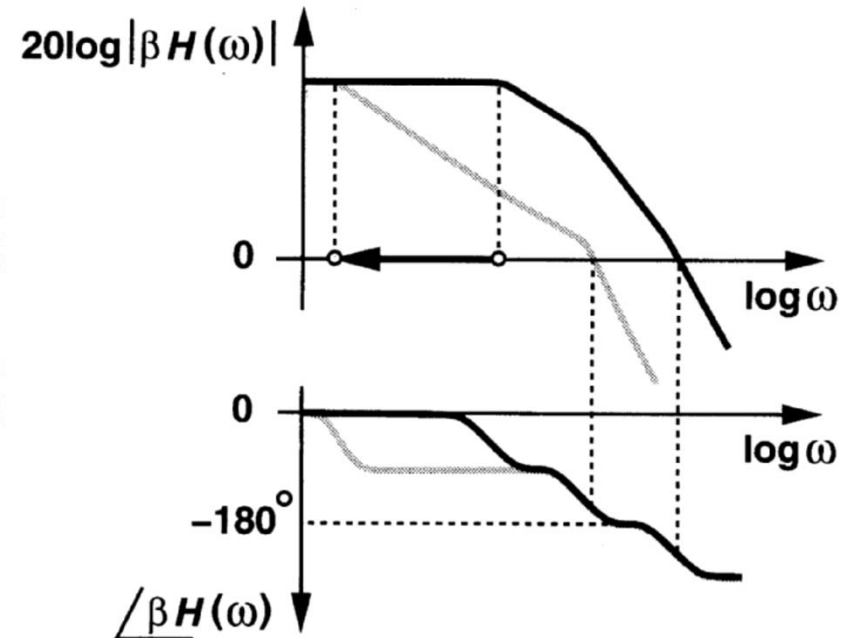
Compensation: increasing C_L

Without compensation



unstable

Translating the dominant pole towards origin



With compensation

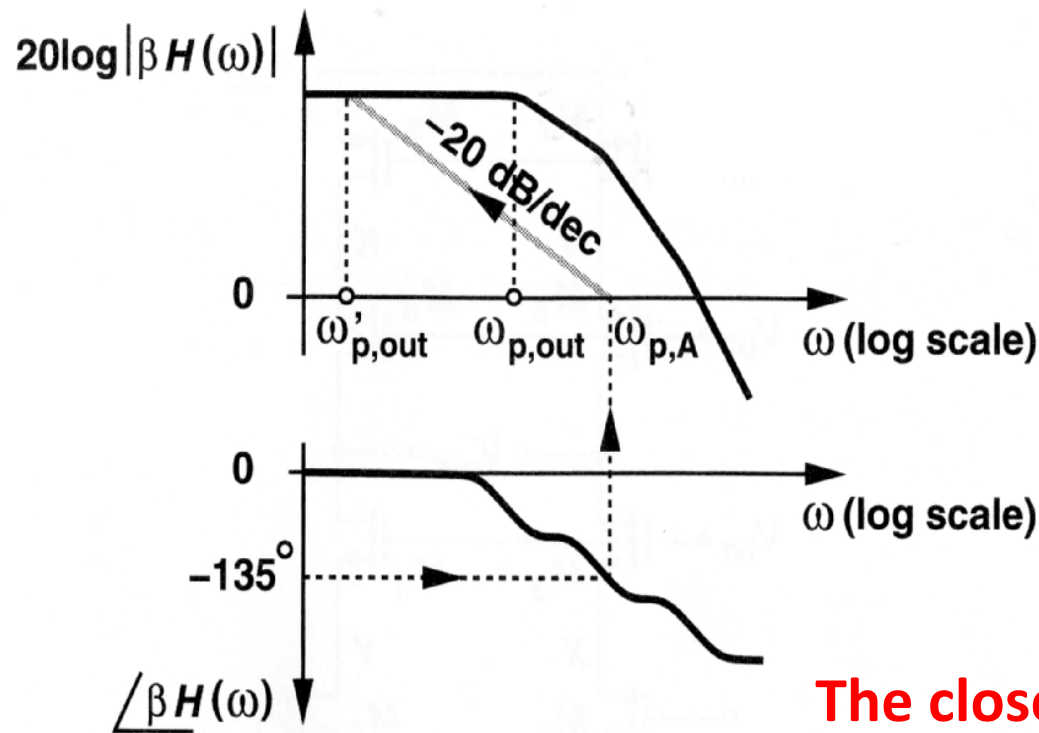


Compensation for a given PM

Example: for 45° phase margin

The loop gain must drop to 0dB at $\omega_{p,A}$

The slope is -20dB/dec



$$\omega_{out} = 1/(r_{out} C_L)$$

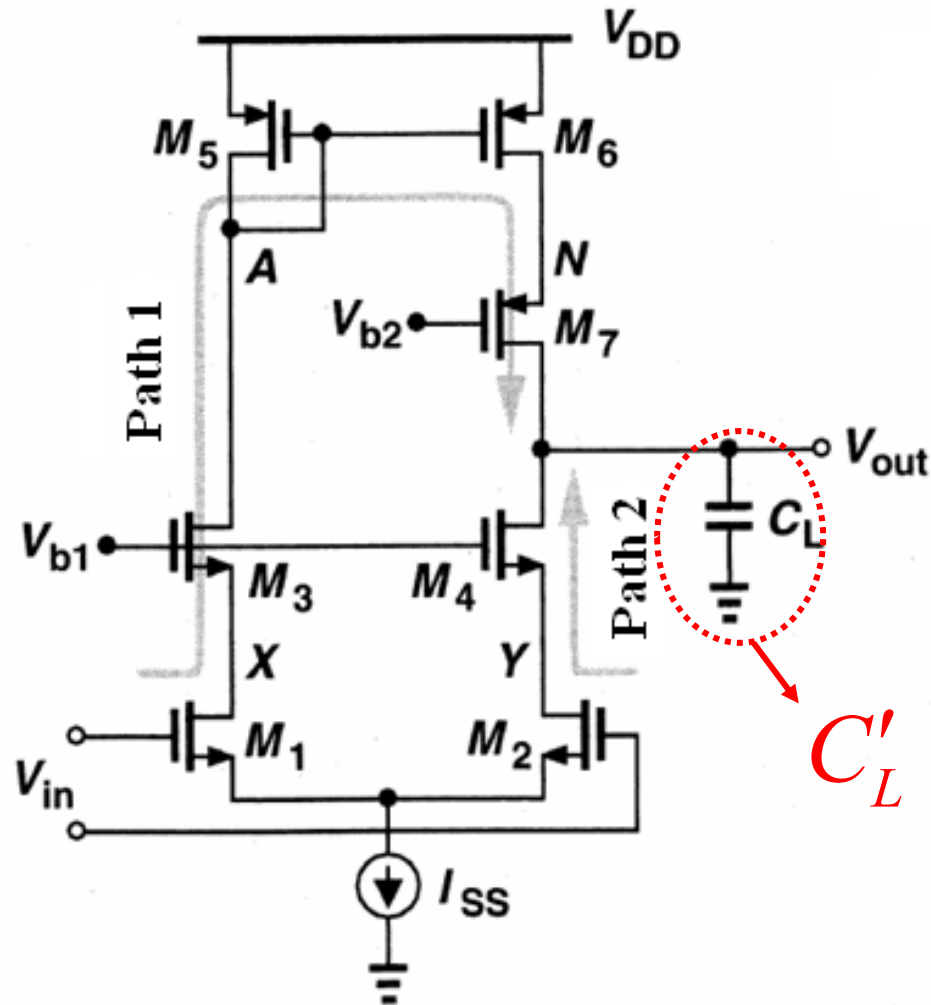
$$\omega'_{out} = 1/(r_{out} C'_L)$$

$$C'_L = \frac{\omega_{out}}{\omega'_{out}} \cdot C_L$$

The closed-loop bandwidth is $\omega_{p,A}$

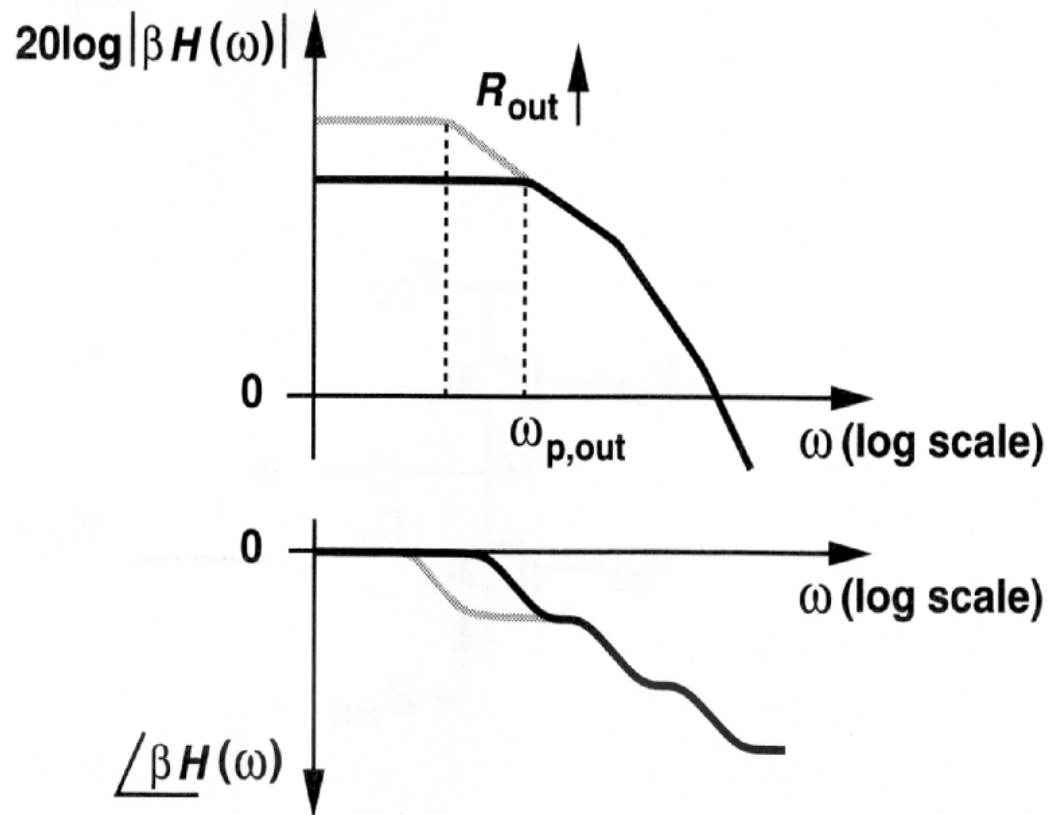


Compensation: increasing the load cap.



Increasing R_{out} can not realize compensation

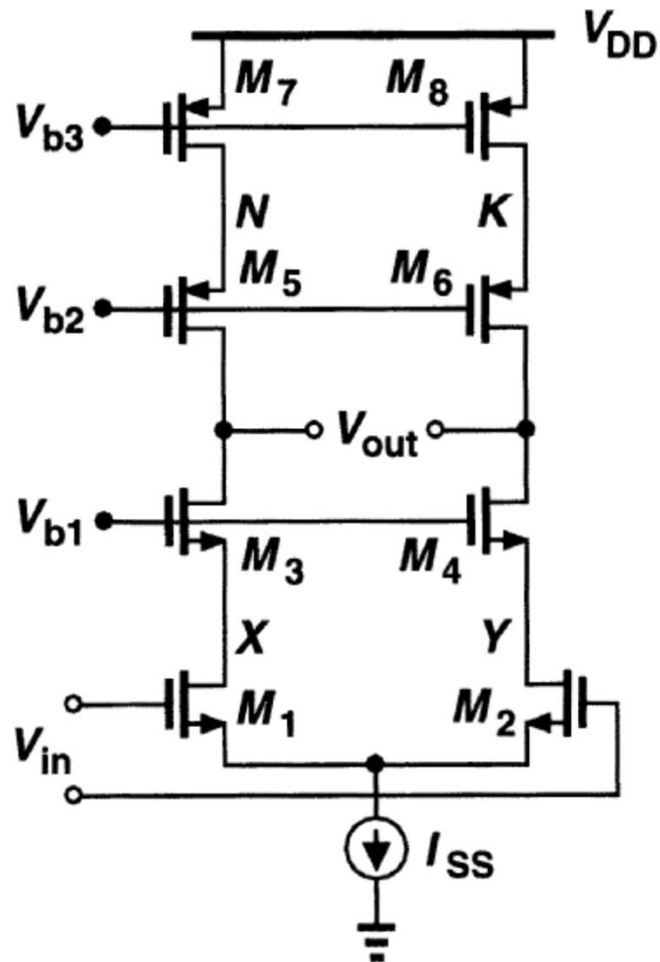
Why?



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 - **Fully differential telescopic OTA**
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Fully Differential Telescopic Op-Amp



Dominant pole is ω_{out}

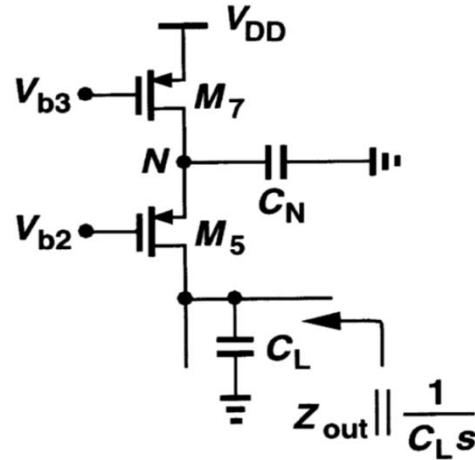
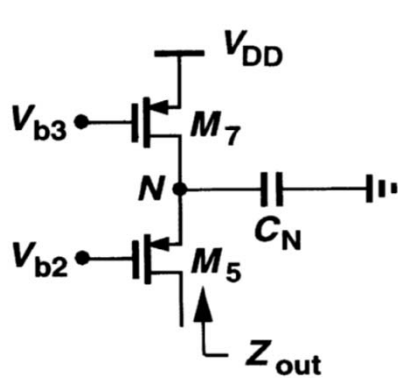
Without mirror pole

The first non-dominant pole is at ω_x

The pole at N/K node can be merged into the output pole



The influence of C_N



C_N lowers ω_{out} by a slight amount

$$Z_{out} \parallel \frac{1}{sC_L} = \frac{(1 + g_{m5}r_{o5}) \frac{r_{o7}}{1 + sr_{o7}C_N} \frac{1}{sC_L}}{(1 + g_{m5}r_{o5}) \frac{r_{o7}}{1 + sr_{o7}C_N} + \frac{1}{sC_L}}$$

$$= \frac{(1 + g_{m5}r_{o5})r_{o7}}{1 + s[(1 + g_{m5}r_{o5})r_{o7}C_L + r_{o7}C_N]}$$

Fully diff. OTA is much more stable and fast than single-ended OTA



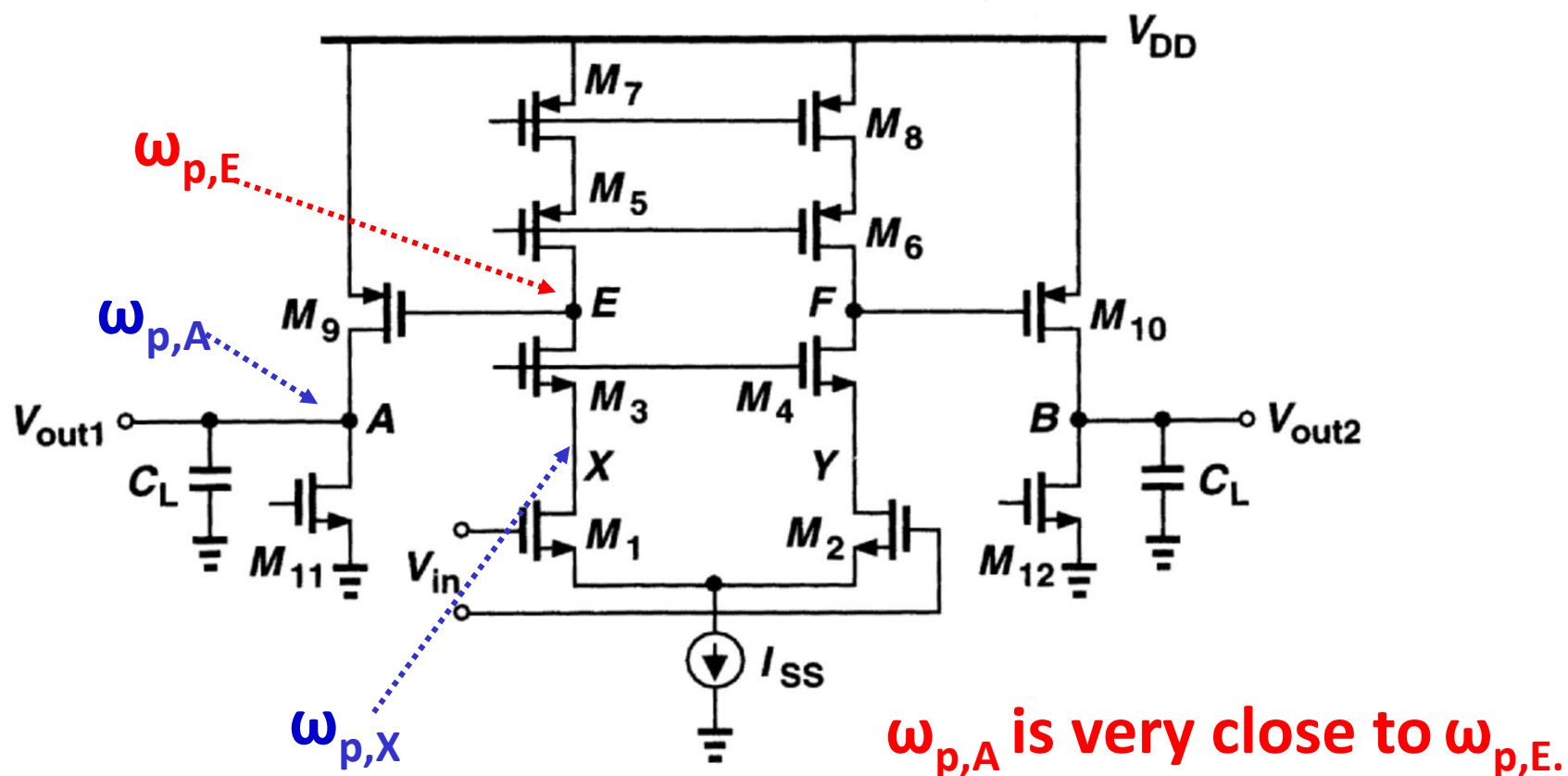
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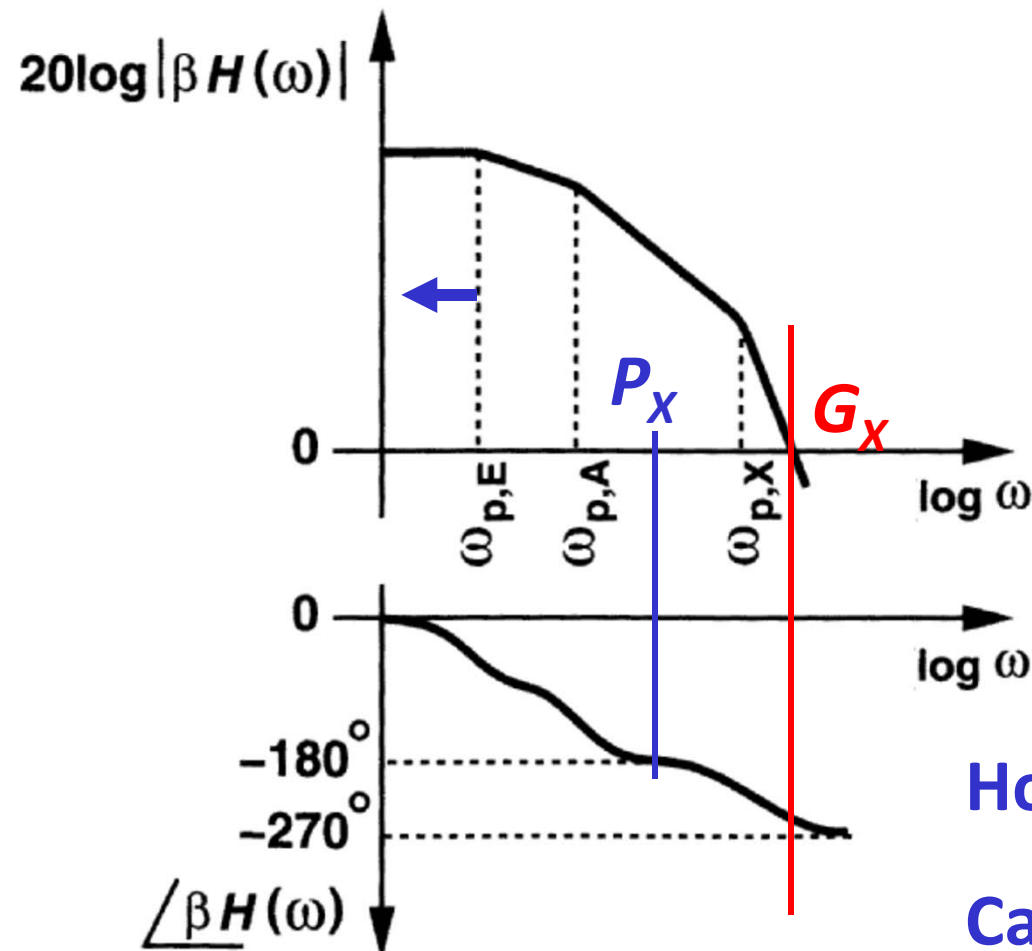


Compensation of Two-Stage Op Amps

High-gain, large –swing two-stage op amp: 3 poles
which is the dominant pole?



Bode plots of two-stage op amp



$P_X < G_X$
Unstable!

How to compensate?

Can we use a large C_L ?



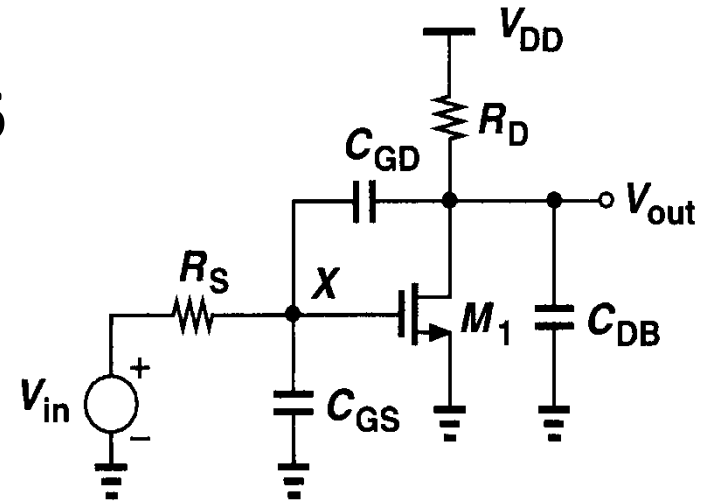
Limitation of compensation by a large C_L

Because $\omega_{p,A}$ is very close to $\omega_{p,E}$.

- If we use a large C_L to move $\omega_{p,E}$ more close to origin
 - A vary large capacitance is needed
 - The available closed-loop bandwidth is limited by $\omega_{p,A}$
- More efficient compensation technique needed
 - Miller compensation
 - *Recall the detail analysis in Chap.6*

Poles for a CS stage with large C_{GD}

- The second stage is a CS stage
 - Recall the analysis in Chapter. 6
 - For large C_{GD}
 - Short at high frequency

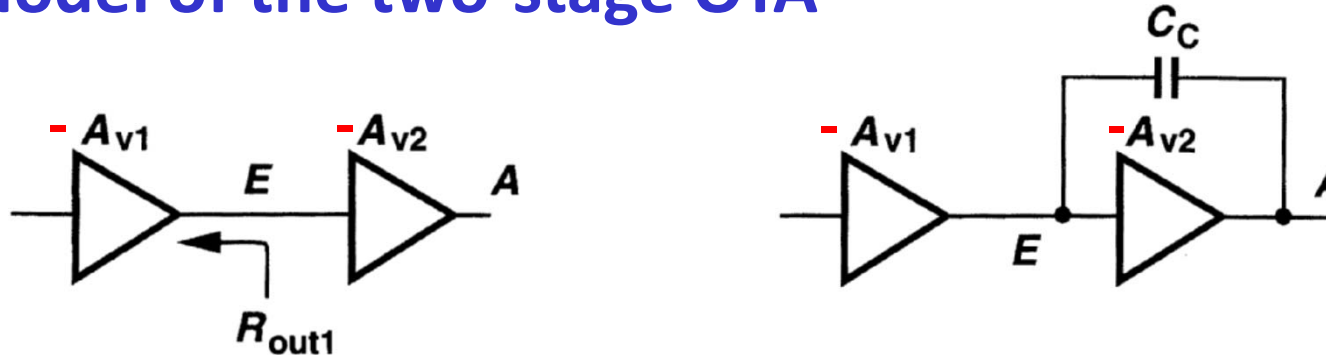


$$\omega_{p1} = \frac{1}{R_S [C_{GS} + (1 + g_m R_D) C_{GD}] + R_D (C_{GD} + C_{DB})}$$

$$\omega_{p2} \approx \frac{g_m}{C_{GS} + C_{DB}}$$

Miller Compensation cont

Model of the two-stage OTA



Miller Effect $C_{eq} = C_E + (1 + A_{v2})C_C$

$$\omega_{pE} = \frac{1}{R_{out1} [C_E + (1 + A_{v2})C_C]}$$

A low frequency pole at the 1st stage is obtained



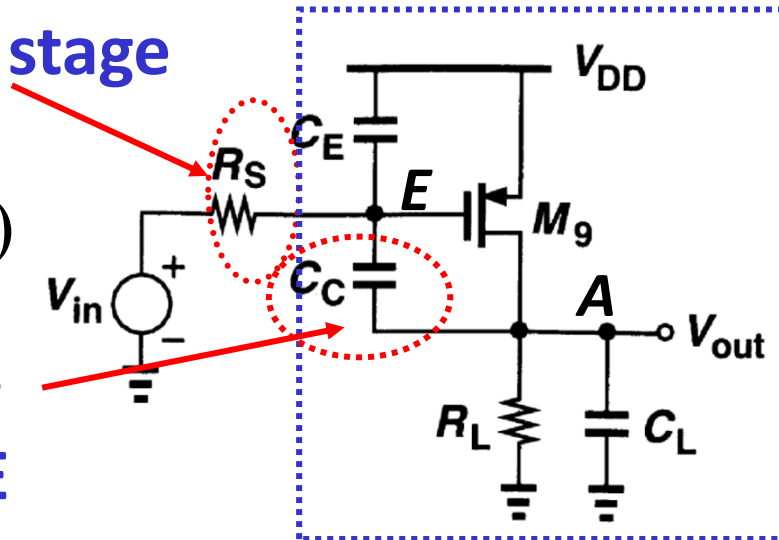
Application of miller compensation to the OTA

Output resistance of the 1st stage

$$R_S = (g_{m6}r_{o6} \cdot r_{o8}) \parallel (g_{m4}r_{o4} \cdot r_{o2})$$

C_C is a large compensation cap.

C_E is the parasitic cap at node E



The 2nd stage

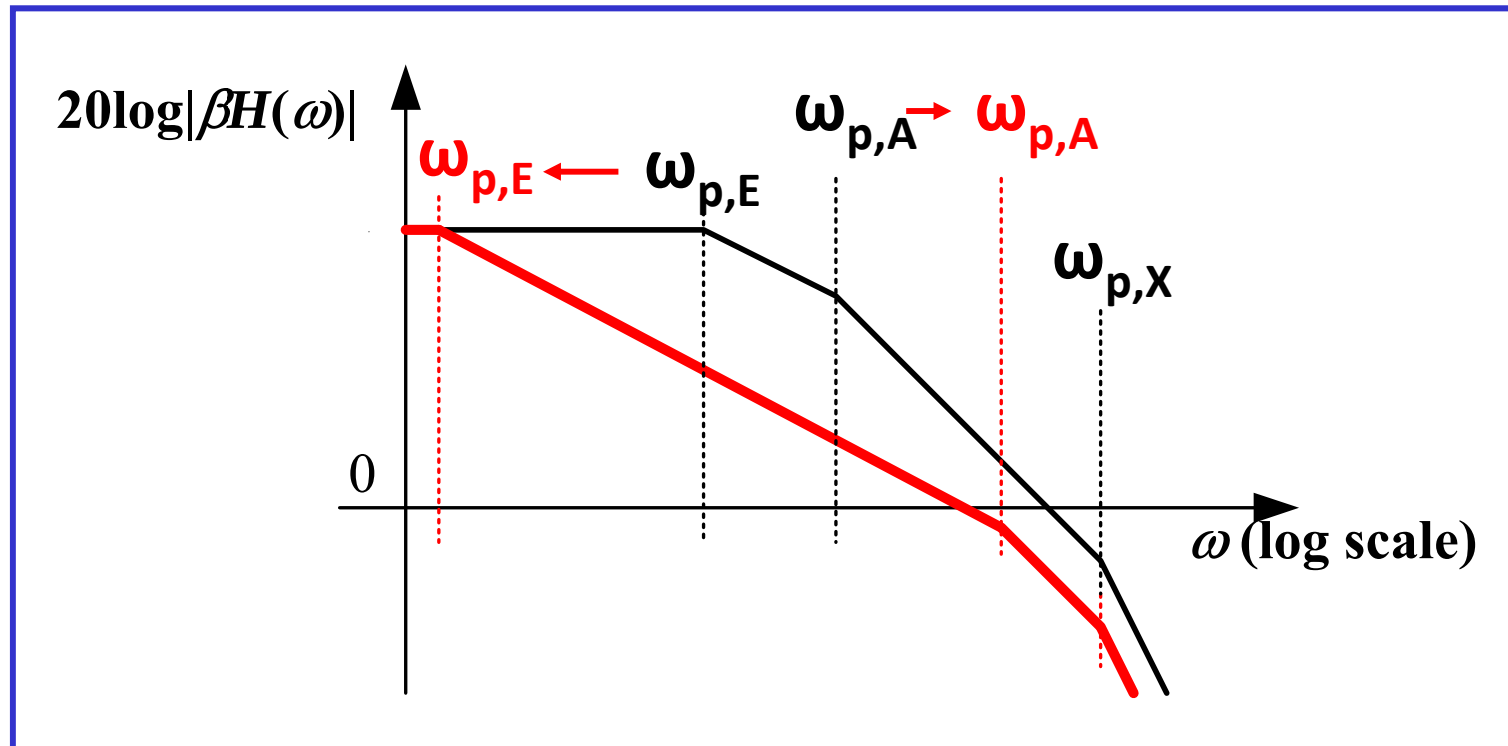
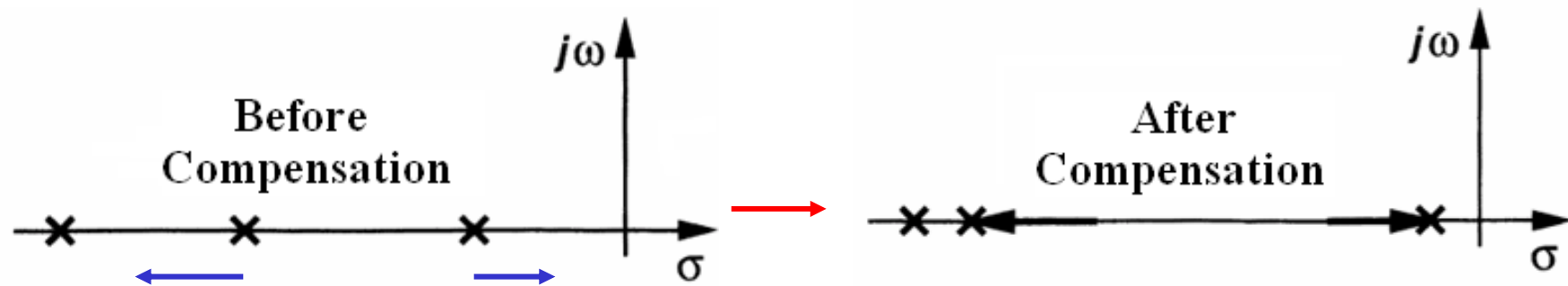
$$\omega_{p,E} = \frac{1}{R_S [C_E + (1 + g_m R_L) C_C] + R_L (C_C + C_L)}$$

$$\approx \frac{1}{R_S [C_E + (1 + g_m R_L) C_C]}$$

$$\omega_{p,A} \approx \frac{g_{m9}}{C_L}$$

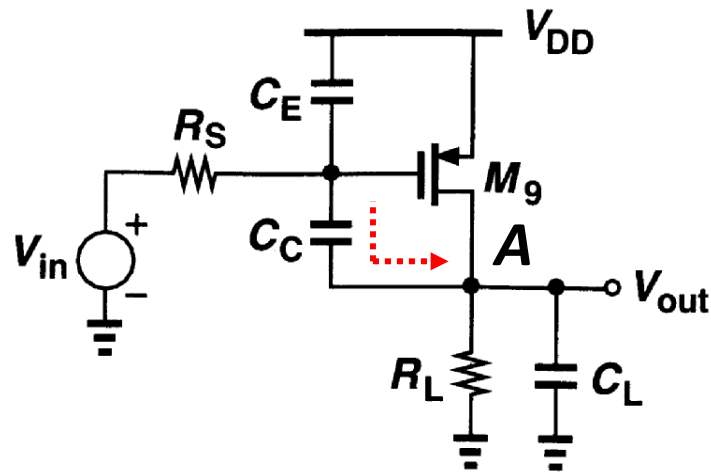
For large C_C , and $C_L \gg C_E$

Pole splitting by the compensation (cont.)



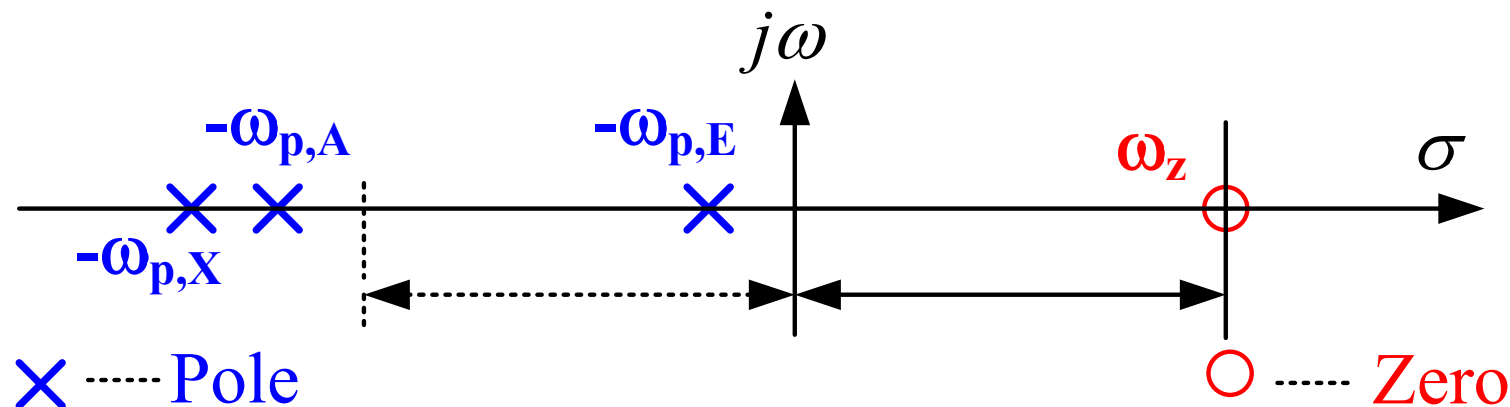
The RHP zero

the feedforward path introduce a RHP zero

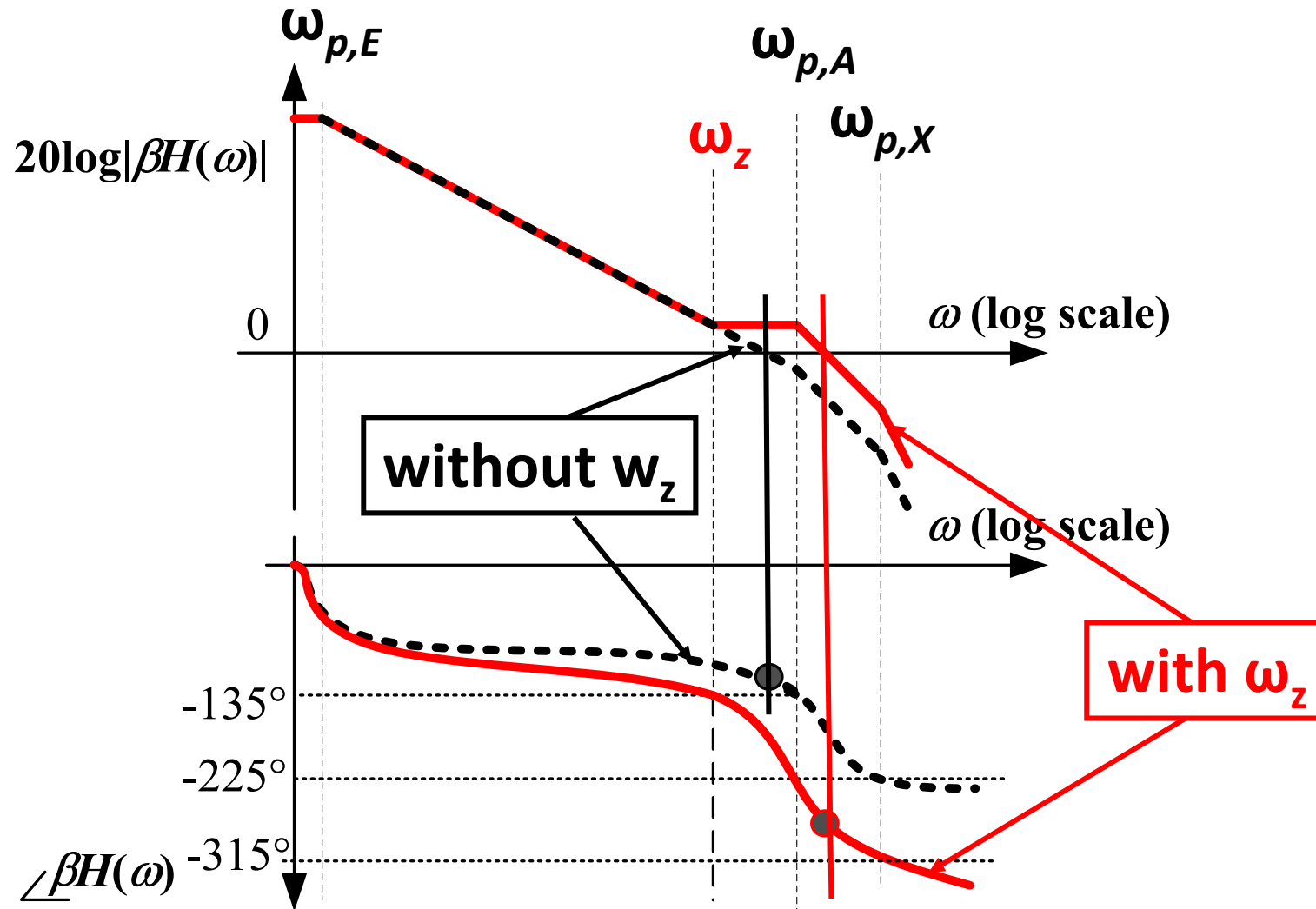


$$\omega_z \approx \frac{g_{m9}}{C_C}$$

$$|\omega_{p,E}| < |\omega_z| < |\omega_{p,A}|$$

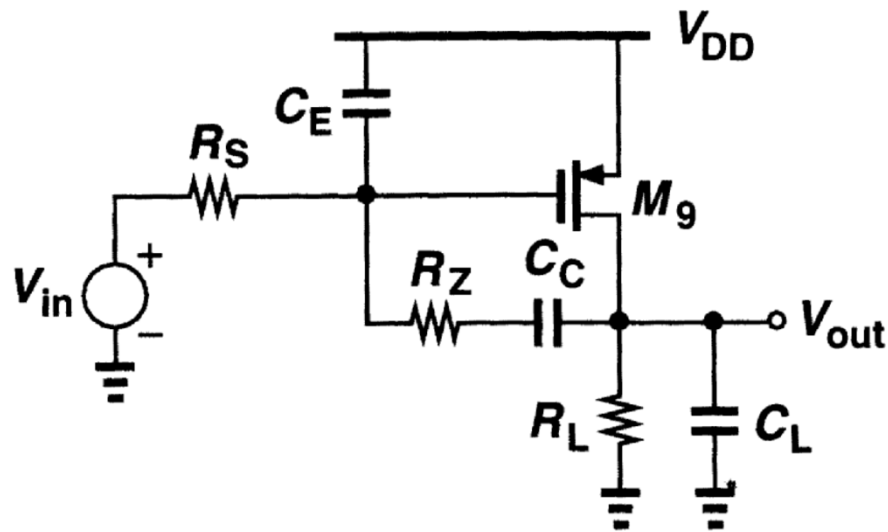


Effect of the RHP zero after compensation



Removal of RHP Zero

Adding R_Z to move the RHP zero to high frequency



$$\frac{1}{R_Z + \frac{1}{s_z C_C}} = g_{m9}$$

$$s_z C_C = s_z R_Z C_C g_{m9} + g_{m9}$$

$$s_z = \frac{g_{m9}}{C_C (1 - R_Z g_{m9})} = \frac{1}{C_C (g_{m9}^{-1} - R_Z)}$$

If $1/g_{m9} = R_Z$,
The RHP zero is disappeared!

$$\rightarrow \omega_z = s_z = \frac{1}{C_C (g_{m9}^{-1} - R_Z)}$$



Calculation of R_Z

- Option I: Get rid of the RHP zero

$$R_Z = 1 / g_{m9}$$

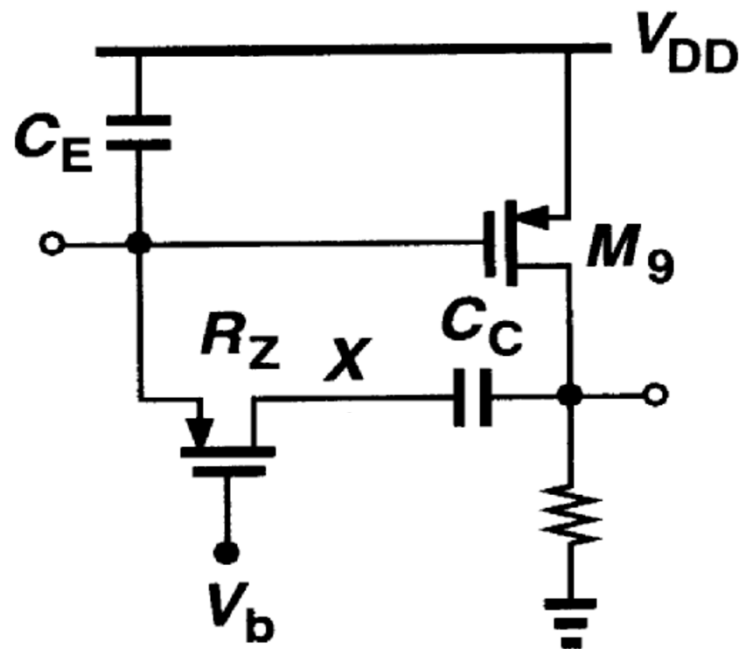
- Option II: move ω_z to LHP, and cancel $\omega_{p,A}$

$$\omega_z = \frac{1}{C_C (1 / g_{m9} - R_Z)}$$
$$\omega_{p,A} = \frac{-g_{m9}}{C_L}$$
$$\frac{1}{C_C (1 / g_{m9} - R_Z)} = \frac{-g_{m9}}{C_L}$$
$$R_Z = \frac{C_L + C_C}{g_{m9} C_C}$$



Realization of R_Z

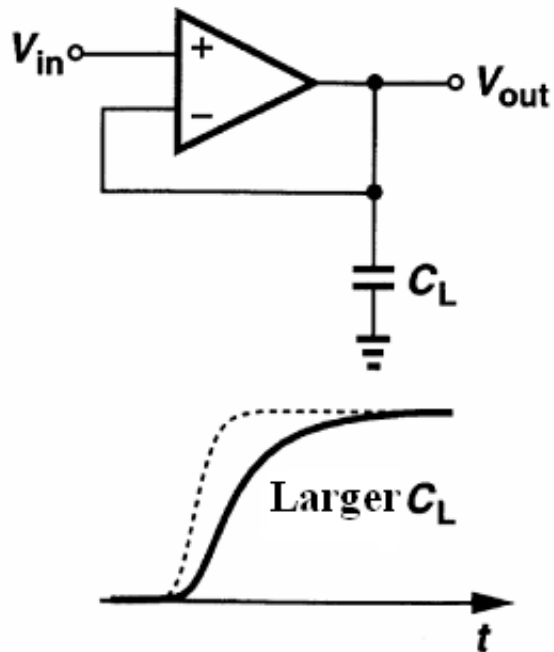
Use a MOS in linear region to obtain an adjustable R_Z



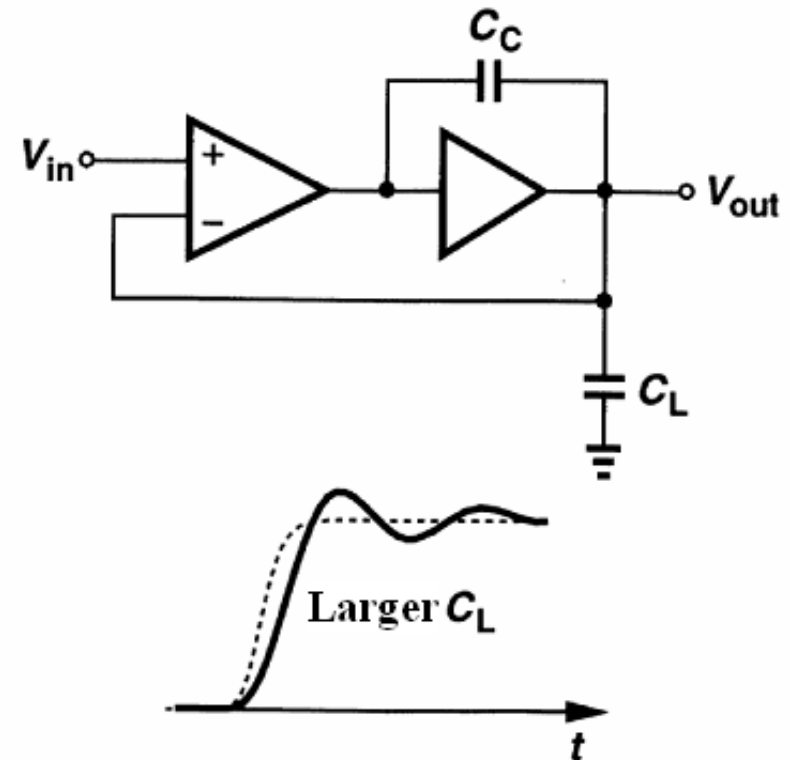
Load Capacitance Effects comparison

Step response comparison

single-stage



Two-stage with Miller



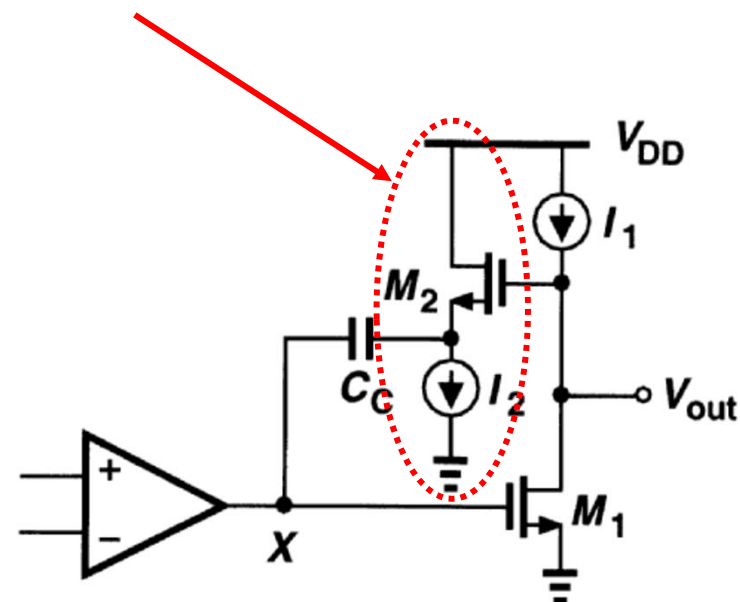
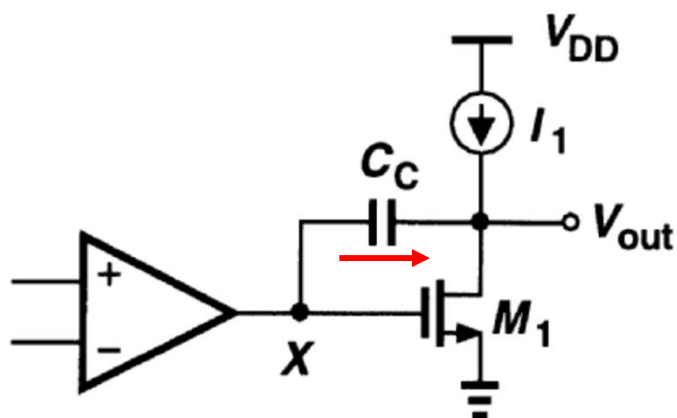
Outline

- General consideration
- Multipole systems
- Phase Margin
- Frequency compensation
- Compensation of two-stage OTAs
- **Other compensation techniques**

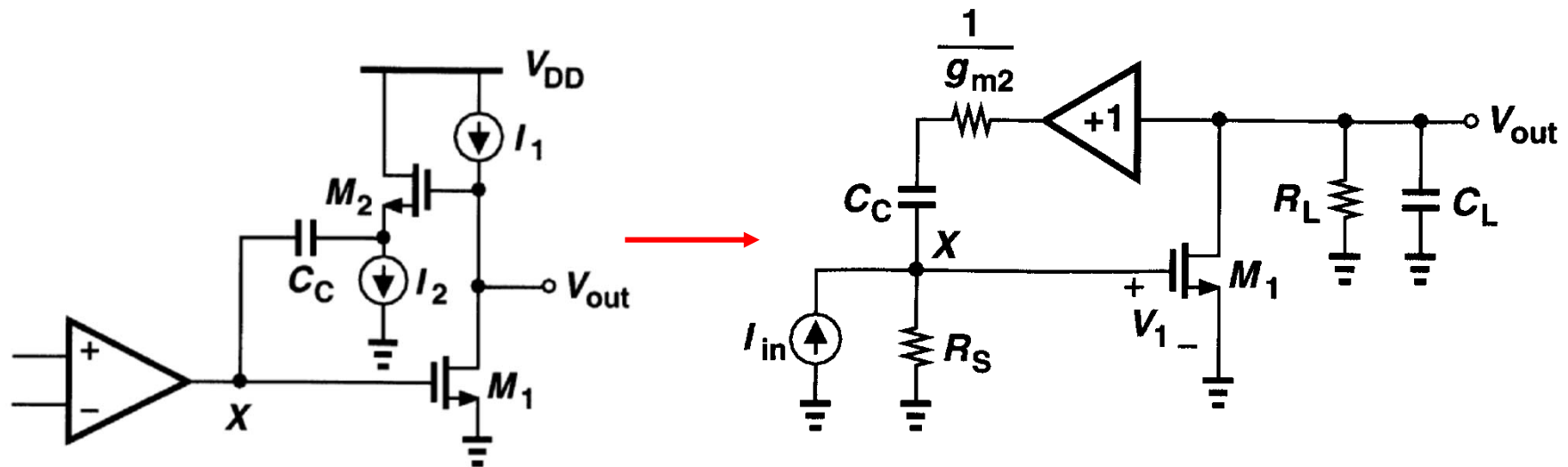


Removal of the RHP zero (1)

Using a source follower to cut the feedforward path



Calculation of the poles



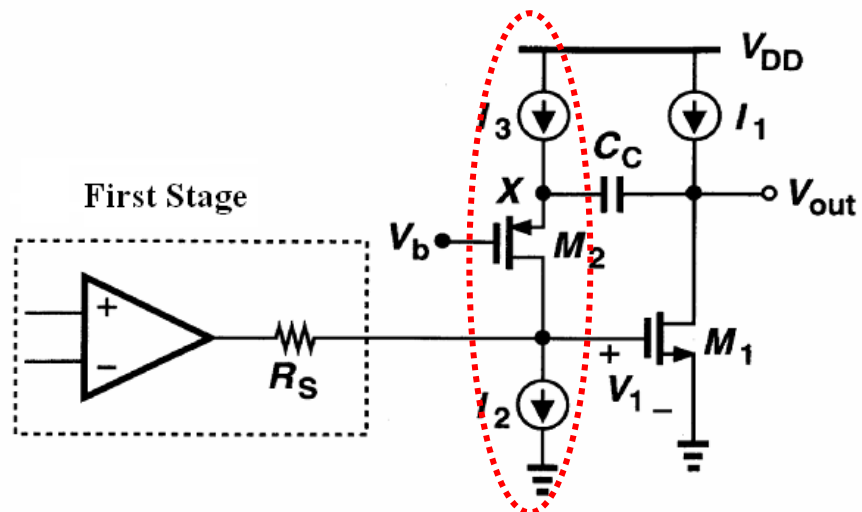
$$\omega_{p1} \approx \frac{1}{R_S g_{m1} R_L C_C}$$

$$\omega_{p2} \approx \frac{g_{m1}}{C_L}$$

No RHP zero anymore!

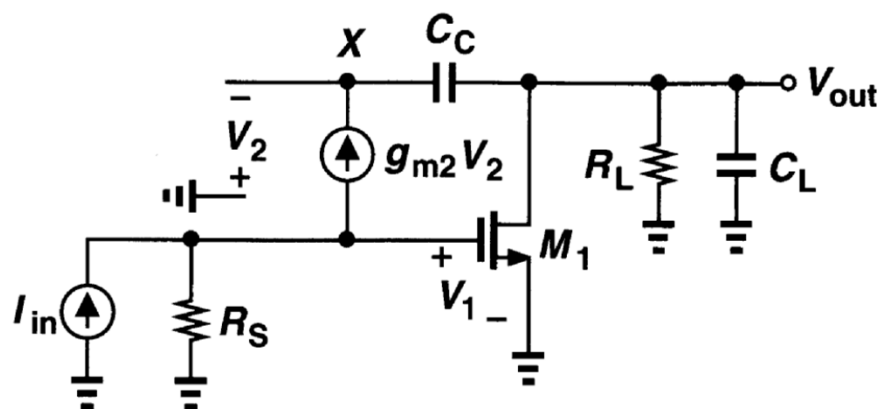
Removal of the RHP zero (2)

Compensation technique using a common-gate stage



$$\omega_{p1} \approx \frac{1}{R_S g_{m1} R_L C_C}$$

$$\omega_{p2} \approx \frac{g_{m2} R_S g_{m1}}{C_L}$$

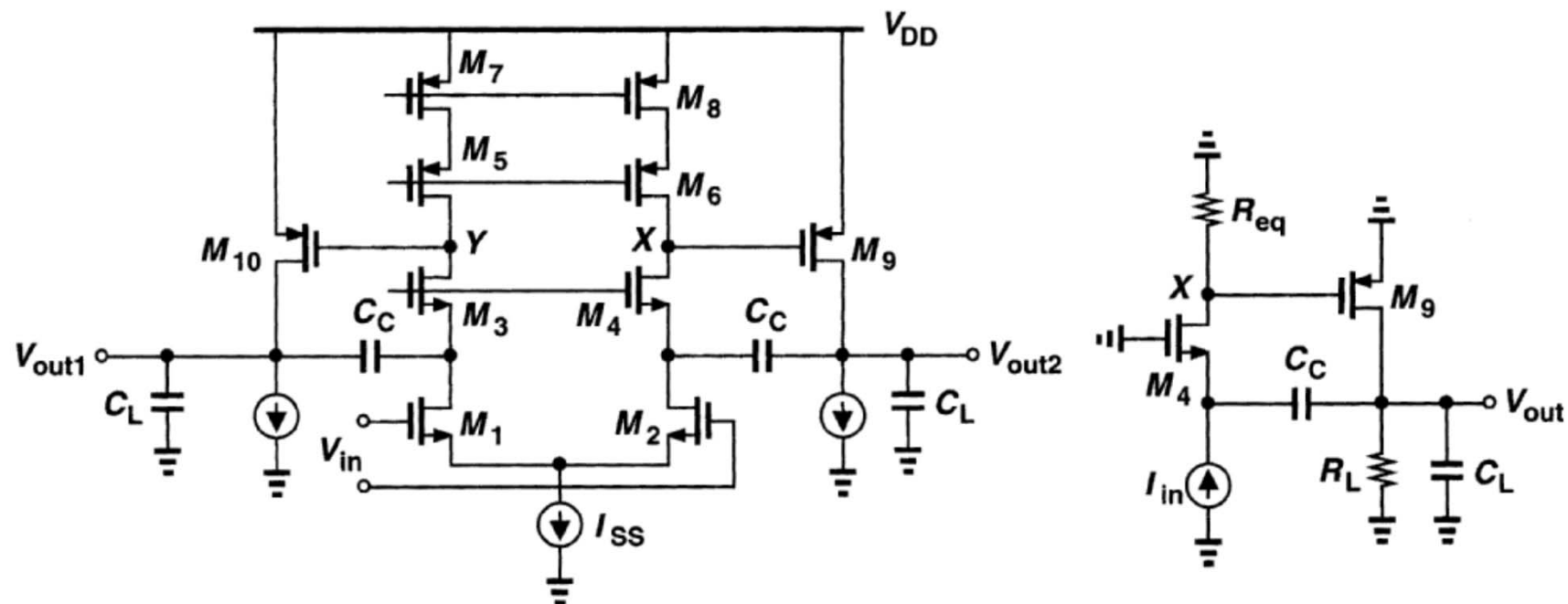


Removal of the RHP zero (3)

Take advantage of the cascode topology

The feedback is connected to the source of M4/M3

Cutting the feedforward path



summary

- General consideration
 - Barkhausen criteria
- Multipole systems
- Phase Margin
- Frequency compensation
 - Single-ended telescopic OTA
 - Fully differential telescopic OTA
- Compensation of two-stage OTAs
 - Miller compensation
 - Removal of RHP zero
- Other compensation techniques